nature medicine

Article

https://doi.org/10.1038/s41591-025-03566-1

Pembrolizumab plus chemotherapy in advanced or recurrent endometrial cancer: overall survival and exploratory analyses of the NRG GY018 phase 3 randomized trial

Received: 23 September 2024

Accepted: 5 February 2025

Published online: 5 March 2025

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Historically, the treatment of patients with advanced stage or recurrent endometrial cancer included paclitaxel plus carboplatin. Immunotherapy in combination with chemotherapy resulted in improved clinical outcomes in several solid tumors. In the phase 3 NRG GY018 study, pembrolizumab plus chemotherapy significantly improved investigator-assessed progression-free survival (PFS; primary endpoint) versus placebo plus chemotherapy in patients with advanced/metastatic/recurrent endometrial cancer regardless of mismatch repair status. Here we report on key secondary endpoints and exploratory analyses. Patients were women ≥18 years old with newly diagnosed stage III or IVA endometrial cancer with measurable disease, or stage IVB or recurrent endometrial cancer with or without measurable disease. Patients (n = 810) were randomized (1:1) to pembrolizumab or placebo plus paclitaxel–carboplatin followed by maintenance pembrolizumab or placebo for up to 24 months. Overall survival was a secondary endpoint and PFS per RECIST v.1.1 by blinded independent central review was an exploratory endpoint. Overall survival data were immature; hazard ratios favored pembrolizumab (mismatch repair-proficient: 0.79 (0.53–1.17); 1-sided nominal P = 0.1157; mismatch repair-deficient: 0.55 (0.25–1.19); 1-sided nominal P = 0.0617). Hazard ratios (95% confidence intervals) for PFS per blinded independent central review favored pembrolizumab (mismatch repair-proficient: 0.64 (0.49–0.85); P = 0.0008; mismatch repair-deficient: 0.45 (0.27–0.73); P = 0.0005). These findings further support the use of pembrolizumab plus chemotherapy as first-line treatment for patients with advanced stage or recurrent endometrial cancer regardless of mismatch repair status. ClinicalTrials.gov identifier: NCT03914612.

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Numerous studies have investigated the efficacy and safety of immune checkpoint inhibitors, alone and in combination with other agents, for the treatment of patients with advanced or recurrent endometrial cancer[1,2]. In the phase 2 KEYNOTE-158 study, monotherapy with the anti-PD-1 monoclonal antibody pembrolizumab resulted in objective responses in 48% (95% confidence interval (CI), 37–60%) of patients with previously treated advanced, microsatellite instability-high or mismatch repair-deficient (dMMR) endometrial cancer[3]. In the phase 3 KEYNOTE-775 study, combination therapy with pembrolizumab and the multitargeted tyrosine kinase inhibitor lenvatinib led to statistically significant and clinically meaningful improvements in progression-free survival (PFS) and overall survival (OS) versus chemotherapy of the treating physician's choice (doxorubicin or paclitaxel) in patients with previously treated advanced, recurrent or metastatic endometrial cancer[4,5]. Benefits were seen in patients with mismatch repair-proficient (pMMR) tumors and all-comers. Results from the KEYNOTE-158 and KEYNOTE-775 studies led to regulatory approval of pembrolizumab alone (dMMR or tumor mutational burden-high) or with lenvatinib (pMMR or all-comers, depending on country) in patients with endometrial cancer following prior systemic platinum-based therapy[6,7].

The phase 3 NRG GY018 study evaluated the addition of pembrolizumab to first-line paclitaxel plus carboplatin in patients with advanced, metastatic or recurrent endometrial cancer[8]. Pembrolizumab plus paclitaxel and carboplatin resulted in statistically significant and clinically meaningful improvement in investigator-assessed PFS (primary endpoint) versus placebo plus chemotherapy in both the pMMR and dMMR populations at interim analysis (IA). Pembrolizumab did not increase the frequency of adverse events (AEs) commonly associated with paclitaxel plus carboplatin[8]. Based on these results, the pembrolizumab plus chemotherapy regimen was approved, irrespective of mismatch repair (MMR) status, in the United States (17 June 2024) and Europe (24 October 2024) and is now included as a recommended option in several treatment guidelines[9,10]. We present secondary and exploratory analyses from the NRG GY018 study.

Results

Interim analyses

Patients. At IA (data cutoff (DCO)), 6 December 2022 for pMMR; 16 December 2022 for dMMR), 810 patients were randomized across four countries (United States, Canada, Japan and Republic of Korea). In the pMMR population (n = 588), 294 patients were randomized to pembrolizumab plus chemotherapy and 294 were randomized to placebo plus chemotherapy (Fig. 1). In the dMMR population (n = 222), 110 patients were randomized to pembrolizumab plus chemotherapy and 112 were randomized to placebo plus chemotherapy. Demographics and baseline clinical characteristics were generally balanced between treatment groups in both the pMMR and dMMR populations (Table 1). Most patients in both populations had PD-L1 combined positive score (CPS) ≥ 1 tumor (71.7%, pMMR; 83.6%, dMMR). Median (range) follow-up for IA was 10.0 (0.0–37.8) months in the pMMR population and 14.4 (4.0–39.4) months in the dMMR population.

Primary endpoint. *PFS by investigator assessment*. Using the intention-to-treat principle, in the pMMR population, 95 events (32.3%) occurred in the pembrolizumab plus chemotherapy group and 138 (46.9%) occurred in the placebo plus chemotherapy group (Fig. 2a). Median PFS was 13.1 (95% CI, 10.6–19.5) and 8.7 (95% CI, 8.4–11.0) months, respectively. The hazard ratio (HR) for PFS significantly favored pembrolizumab plus chemotherapy (0.57 (95% CI, 0.44–0.74); $P < 0.0001$ (prespecified boundary for statistical significance, 0.001162)). In the dMMR population, 29 events (26.4%) occurred in the pembrolizumab plus chemotherapy group and 60 (53.6%) occurred in the placebo plus chemotherapy group (Fig. 2b). Median PFS was not reached and 8.3 (95% CI, 6.5–12.3) months, respectively. The HR for PFS significantly favored pembrolizumab plus chemotherapy (0.34 (95% CI, 0.22–0.53); $P < 0.0001$ (prespecified boundary for statistical significance, 0.002074)).

Secondary endpoints. *Local and central MMR immunohistochemistry*. There was a high level of concordance between local and central MMR immunohistochemistry (IHC) assessments (Supplementary Information and Supplementary Table 1). Among 749 patients with both local and central results, 721 (96.3%) had concordant MMR status; Cohen kappa coefficient was 0.9068.

Overall survival. Survival data were immature at IA. The pMMR population had an information fraction of 27.2%; 45 deaths (15.3%) occurred in the pembrolizumab plus chemotherapy group and

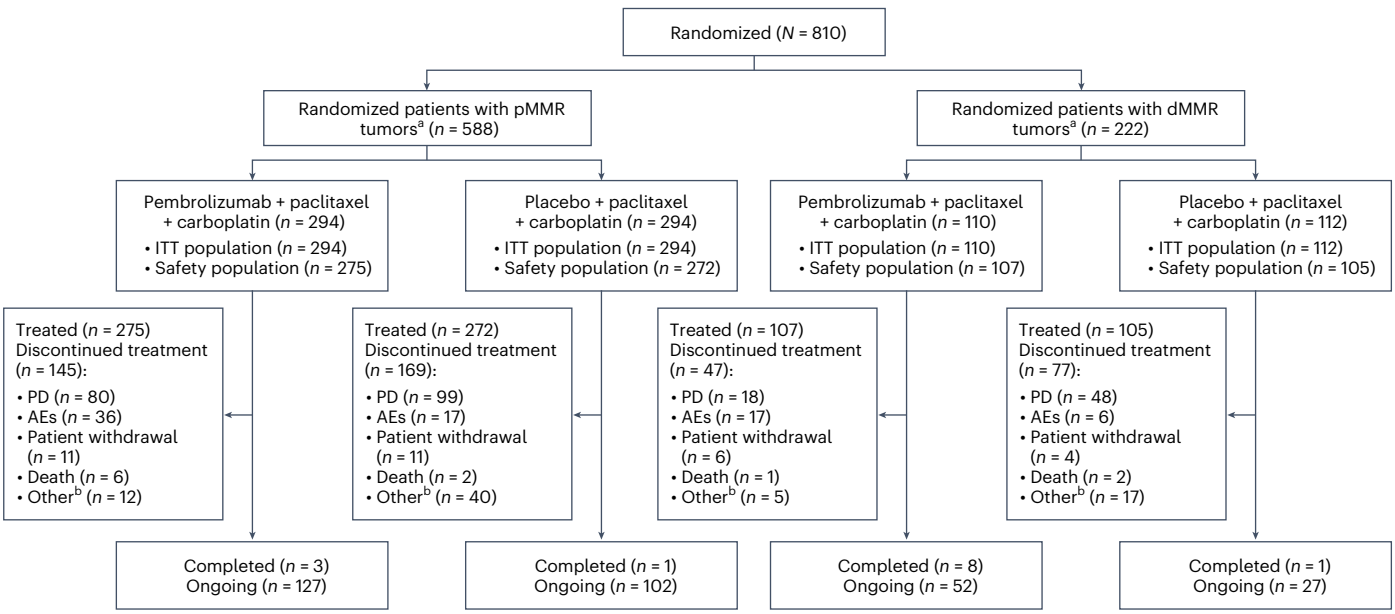


Fig. 1 | Patient disposition in the pMMR and dMMR populations at interim analysis. Data cutoff dates: 6 December 2022 (pMMR); 16 December 2022 (dMMR). [a]By central or local immunohistochemistry. [b]Includes alternative therapy (in absence of progression), off treatment due to other complicating disease, symptomatic deterioration and other (not otherwise specified). PD, progressive disease.

https://doi.org/10.1038/s41591-025-03566-1

Table 1 | Demographics and baseline clinical characteristics at interim analysis (ITT population)

	pMMR population			dMMR population		
	Pembrolizumab + chemotherapy (*n*=294)	Placebo + chemotherapy (*n*=294)	All patients (*n*=588)	Pembrolizumab + chemotherapy (*n*=110)	Placebo + chemotherapy (*n*=112)	All patients (*n*=222)
Age, median (range), years	66.0 (31.0–94.0)	66.1 (29.0–91.0)	66.1 (29.0–94.0)	67.2 (39.0–82.0)	66.0 (37.0–86.0)	66.1 (37.0–86.0)
Race, *n* (%)						
White	212 (72.1)	212 (72.1)	424 (72.1)	91 (82.7)	85 (75.9)	176 (79.3)
Black or African American	46 (15.6)	50 (17.0)	96 (16.3)	10 (9.1)	9 (8.0)	19 (8.6)
Asian	17 (5.8)	14 (4.8)	31 (5.3)	3 (2.7)	4 (3.6)	7 (3.2)
American Indian or Alaska Native	2 (0.7)	2 (0.7)	4 (0.7)	0 (0)	2 (1.8)	2 (0.9)
Native Hawaiian or other Pacific Islander	1 (0.3)	3 (1.0)	4 (0.7)	0 (0)	0 (0)	0 (0)
Multiracial	1 (0.3)	1 (0.3)	2 (0.3)	0 (0)	0 (0)	0 (0)
Missing data	15 (5.1)	12 (4.1)	27 (4.6)	6 (5.5)	12 (10.7)	18 (8.1)
Hispanic ethnic group, *n* (%)						
No	265 (90.1)	274 (93.2)	539 (91.7)	104 (94.5)	97 (86.6)	201 (90.5)
Yes	21 (7.1)	14 (4.8)	35 (6.0)	4 (3.6)	7 (6.3)	11 (5.0)
Unknown	4 (1.4)	3 (1.0)	7 (1.2)	1 (0.9)	4 (3.6)	5 (2.3)
Not reported	4 (1.4)	3 (1.0)	7 (1.2)	1 (0.9)	4 (3.6)	5 (2.3)
ECOG PS, *n* (%)						
0	198 (67.3)	197 (67.0)	395 (67.2)	70 (63.6)	72 (64.3)	142 (64.0)
1	87 (29.6)	88 (29.9)	175 (29.8)	39 (35.5)	35 (31.3)	74 (33.3)
2	9 (3.1)	9 (3.1)	18 (3.1)	1 (0.9)	5 (4.5)	6 (2.7)
PD-L1 CPS,[a] *n*	288	288	576	108	111	219
≥1, *n* (%)	208 (72.2)	205 (71.2)	413 (71.7)	86 (79.6)	97 (87.4)	183 (83.6)
<1, *n* (%)	80 (27.8)	83 (28.8)	163 (28.3)	22 (20.4)	14 (12.6)	36 (16.4)
Methylation, *n*	–	–	–	94	94	188
Yes, *n* (%)	–	–	–	81 (86.2)	77 (81.9)	158 (84.0)
No, *n* (%)	–	–	–	13 (13.8)	17 (18.1)	30 (16.0)
Histology, *n* (%)						
Adenocarcinoma NOS	24 (8.2)	35 (11.9)	59 (10.0)	12 (10.9)	12 (10.7)	24 (10.8)
Clear cell	19 (6.5)	20 (6.8)	39 (6.6)	0 (0)	0 (0)	0 (0)
Dedifferentiated or undifferentiated	7 (2.4)	6 (2.0)	13 (2.2)	4 (3.6)	4 (3.6)	8 (3.6)
Endometrioid						
Grade 1	55 (18.7)	45 (15.3)	100 (17.0)	20 (18.2)	34 (30.4)	54 (24.3)
Grade 2	51 (17.3)	59 (20.1)	110 (18.7)	52 (47.3)	43 (38.4)	95 (42.8)
Grade 3	53 (18.0)	42 (14.3)	95 (16.2)	15 (13.6)	16 (14.3)	31 (14.0)
Mixed epithelial	6 (2.0)	10 (3.4)	16 (2.7)	3 (2.7)	2 (1.8)	5 (2.3)
Serous	79 (26.9)	76 (25.9)	155 (26.4)	4 (3.6)	1 (0.9)	5 (2.3)
Missing	0 (0)	1 (0.3)	1 (0.2)	0 (0)	0 (0)	0 (0)
FIGO stage at initial diagnosis						
I	102 (34.7)	103 (35.0)	205 (34.9)	44 (40.0)	57 (50.9)	101 (45.5)
II	25 (8.5)	25 (8.5)	50 (8.5)	14 (12.7)	13 (11.6)	27 (12.2)
III	56 (19.0)	46 (15.6)	102 (17.3)	18 (16.4)	11 (9.8)	29 (13.1)
IVA	11 (3.7)	5 (1.7)	16 (2.7)	2 (1.8)	4 (3.6)	6 (2.7)
IVB	100 (34.0)	115 (39.1)	215 (36.6)	32 (29.1)	27 (24.1)	59 (26.6)
Previous therapy, *n* (%)						
Chemotherapy	74 (25.2)	76 (25.9)	150 (25.5)	4 (3.6)	8 (7.1)	12 (5.4)
Radiotherapy	118 (40.1)	124 (42.2)	242 (41.2)	42 (38.2)	54 (48.2)	96 (43.2)
Surgery	265 (90.1)	248 (84.4)	513 (87.2)	97 (88.2)	104 (92.9)	201 (90.5)

Data cutoff dates: 6 December 2022 (pMMR); 16 December 2022 (dMMR). FIGO, International Federation of Gynecology and Obstetrics; ECOG PS, Eastern Cooperative Oncology Group performance status; NOS, not otherwise specified. [a]CPS was defined as the number of PD-L1-positive cells (tumor cells, lymphocytes and macrophages) divided by the total number of tumor cells, multiplied by 100.

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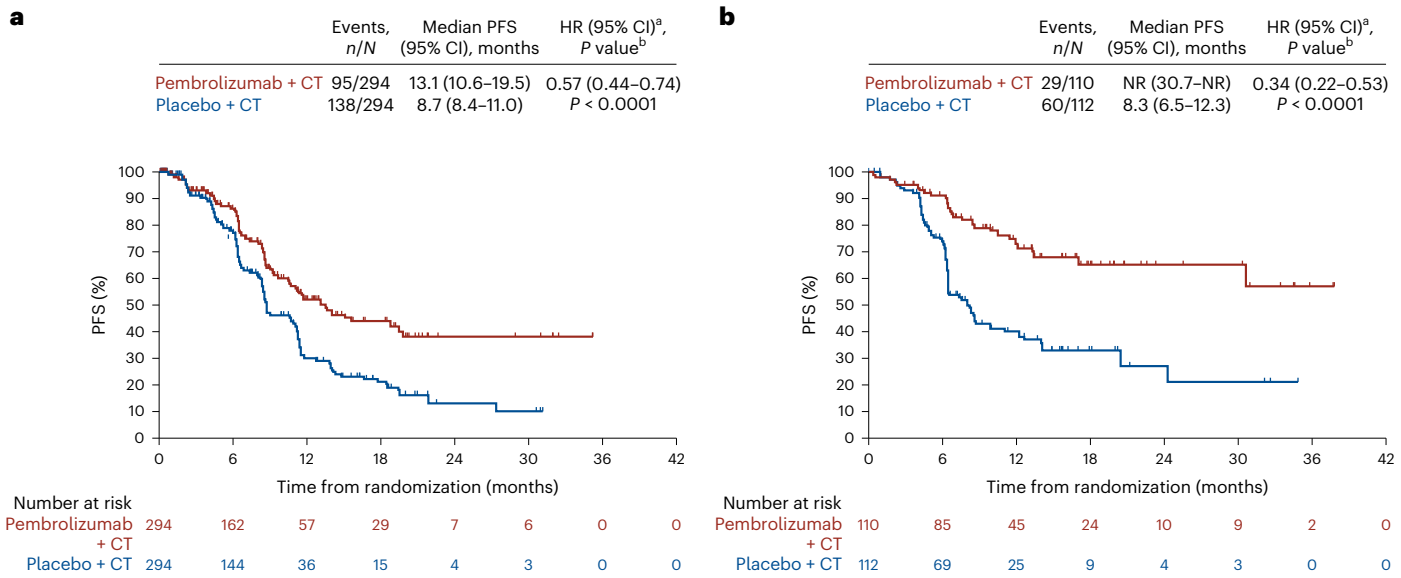


Fig. 2 | Kaplan–Meier-estimated PFS per RECIST v.1.1 as assessed by the investigators at IA. a,**b**, Kaplan–Meier-estimated PFS per RECIST v.1.1 at IA in the pMMR (**a**) and dMMR (**b**) populations. Tick marks indicate censored data. Data cutoff dates: 6 December 2022 (pMMR); 16 December 2022 (dMMR). [a]Based on a Cox regression model with the Efron method of tie handling with treatment as a covariate stratified by prior chemotherapy. [b]One-sided *P* value based on log-rank test stratified by prior chemotherapy. CT, chemotherapy; NR, not reached; pembro, pembrolizumab.

54 (18.4%) occurred in the placebo plus chemotherapy group (Fig. 3a). The HR for OS was 0.79 (95% CI, 0.53–1.17; 1-sided nominal $P = 0.1157$). The dMMR population had an information fraction of 18.0%; 10 deaths (9.1%) occurred in the pembrolizumab plus chemotherapy group and 17 (15.2%) occurred in the placebo plus chemotherapy group (Fig. 3b). The HR for OS was 0.55 (95% CI, 0.25–1.19; 1-sided nominal $P = 0.0617$).

Among those who discontinued treatment, more patients in the placebo versus pembrolizumab group received subsequent PD-1/PD-L1 inhibitors in the pMMR (45.0% (76 of 169) versus 19.3% (28 of 145)) and dMMR (54.5% (42 of 77) versus 19.1% (9 of 47)) populations (Supplementary Information and Supplementary Table 2).

Exploratory endpoints. *PFS by demographics and baseline characteristics*. HRs for PFS by investigator assessment favored pembrolizumab plus chemotherapy across all subgroups defined by demographics and baseline clinical characteristics, including histology, in both the pMMR (Fig. 4a) and dMMR (Fig. 4b) populations.

PFS by tumor PD-L1 expression. In the pMMR population, the HR (95% CI) for PFS by investigator assessment was 0.59 (0.43–0.80) in patients with PD-L1 CPS ≥ 1 tumor (Fig. 4c) and 0.44 (0.26–0.75) in patients with PD-L1 CPS < 1 tumor (Fig. 4d). In the dMMR population, the corresponding HRs (95% CIs) for PFS were 0.27 (0.16–0.47; Fig. 4e) and 0.30 (0.11–0.83; Fig. 4f), respectively, although there were few patients and events in the PD-L1 CPS < 1 subgroup.

PFS by methylation status in dMMR population. The HR (95% CI) for PFS by investigator assessment was 0.30 (0.18–0.50) in patients with methylation and 0.17 (0.04–0.79) in patients without methylation, suggestive of MMR gene mutation (Extended Data Fig. 1).

PFS by blinded independent central review. In the pMMR population, median PFS was 19.5 (95% CI, 13.1–28.0) and 11.0 (95% CI, 9.0–11.5) months with pembrolizumab plus chemotherapy versus placebo plus chemotherapy, respectively (HR, 0.64 (95% CI, 0.49–0.85); 1-sided nominal $P = 0.0008$; Extended Data Fig. 2a). In the dMMR population, median PFS was not reached with pembrolizumab plus chemotherapy and was 14.1 months (95% CI, 8.5 months–not reached) with placebo plus chemotherapy (HR, 0.45 (95% CI, 0.27–0.73); 1-sided nominal $P = 0.0005$; Extended Data Fig. 2b).

Ad hoc analyses (DCO, 18 August 2023)

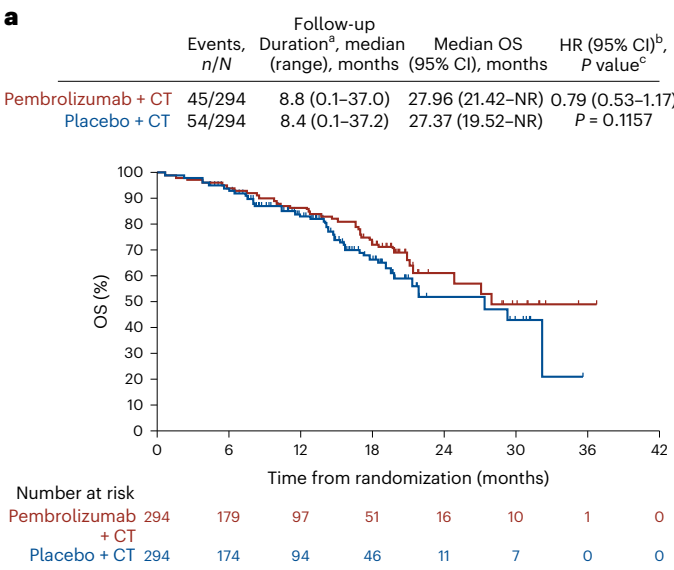
The secondary endpoints of OS, objective response rate (ORR) and safety were assessed at the time of the ad hoc analyses.

Patients. At the time of the ad hoc analysis, 819 patients were randomized (see Extended Data Fig. 3 and Extended Data Table 1 for disposition and demographics and baseline clinical characteristics, respectively). Median (range) follow-up for ad hoc analyses was 20.8 (7.9–46.2) months in the pMMR population and 22.5 (12.0–47.4) months in the dMMR population.

Secondary endpoints. *Overall survival*. The pMMR population had an information fraction of 46.4%; 77 deaths (25.8%) occurred in the pembrolizumab plus chemotherapy group and 92 (30.8%) occurred in the placebo plus chemotherapy group (Extended Data Fig. 4a). Thus, 32 and 38 more OS events, respectively, had occurred at the time of the ad hoc analyses compared with the IA. The HR for OS was 0.80 (95% CI, 0.59–1.08; 1-sided nominal $P = 0.0683$). The dMMR population had an information fraction of 29.3%; 17 deaths (15.5%) occurred in the pembrolizumab plus chemotherapy group and 27 (24.1%) occurred in the placebo plus chemotherapy group (Extended Data Fig. 4b). Thus, 7 and 10 more OS events, respectively, had occurred at the time of the ad hoc analyses compared with the IA. The HR for OS was 0.57 (95% CI, 0.31–1.04; 1-sided nominal $P = 0.0323$).

Among those who discontinued treatment, more patients in the placebo versus pembrolizumab group received subsequent PD-1/PD-L1 inhibitors in the pMMR (42.9% (118 of 275) versus 21.9% (47 of 215)) and dMMR (51.0% (53 of 104) versus 17.9% (10 of 56)) populations (Supplementary Information and Supplementary Table 3). Results of the sensitivity analysis demonstrated a lower HR for OS in the pMMR population (HR, 0.70; 95% CI, 0.50–0.98) when accounting for post-study immunotherapy use (Extended Data Fig. 4c). Due to sparse OS events in the dMMR population with/without subsequent

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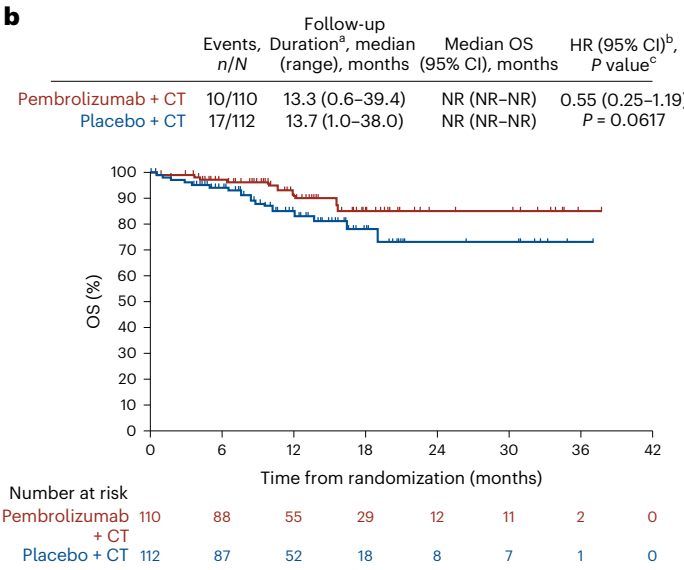


Fig. 3 | Kaplan–Meier-estimated OS at IA. a,**b**, Kaplan–Meier-estimated OS in the pMMR (**a**) and dMMR (**b**) populations at interim analysis. OS data were immature (27.2% information fraction for pMMR population (99 of the 364 events needed for final analysis had occurred) and 18.0% information fraction for dMMR population (27 of the 150 events needed for final analysis had occurred)). Tick marks indicate censored data. Data cutoff dates: 6 December 2022 (pMMR); 16 December 2022 (dMMR). [a]Follow-up duration is the time from randomization to the date of death or the database cutoff date if the participant is still alive. [b]Based on a Cox regression model with the Efron method of tie handling with treatment as a covariate stratified by prior chemotherapy. [c]One-sided *P* value based on log-rank test stratified by prior chemotherapy.

anti-PD-1/PD-L1 therapies with or without lenvatinib, the model and bootstrap samples were not stable for the dMMR population during the first-stage parametric survival model. Consequently, the two-stage model result for the dMMR population is not presented.

Objective response (investigator assessment). The ORR in the pMMR population was 72.3% (95% CI, 66.0%–78.1%) in the pembrolizumab plus chemotherapy group and 59.0% (95% CI, 52.5%–65.3%) in the placebo plus chemotherapy group (Extended Data Fig. 5a and Extended Data Table 2). Median duration of response (DOR) was 8.1 (range, 0.0+ to 40.9+; + indicates no progressive disease at last disease assessment) and 6.4 (range, 0.0+ to 28.3+) months, respectively (Extended Data Fig. 6a). In the dMMR population, the ORR was 82.1% (95% CI, 72.9%–89.2%) in the pembrolizumab plus chemotherapy group and 71.6% (95% CI, 61.4%–80.4%) in the placebo plus chemotherapy group (Extended Data Fig. 5b). Median DOR was not reached (range, 0.0+ to 41.8+) and 4.8 months (range, 0.0+ to 42.2+), respectively (Extended Data Fig. 6b). Waterfall plots are shown in Extended Data Fig. 7. Response rates at IA, when some patients had no post-baseline assessment available for response evaluation, are summarized in Extended Data Table 3. Notably, the ORR and DOR still favored the pembrolizumab plus chemotherapy group.

Safety. In the combined pMMR and dMMR populations, AEs were reported in 99.2% of patients in the pembrolizumab plus chemotherapy group and 99.7% in the placebo plus chemotherapy group (Table 2). Grade ≥3 AEs occurred in 65.7% and 49.2% of patients, respectively. The most common AEs (≥20% of patients in either group) are shown in Table 2. Treatment-related AEs occurred in 96.9% of patients in the pembrolizumab plus chemotherapy group and 96.1% in the placebo plus chemotherapy group and were grade ≥3 in 49.9% and 34.0%, respectively. Study treatment was discontinued because of AEs in 18.2% and 7.2% of patients, respectively (Extended Data Fig. 3). Immune-mediated AEs and infusion reactions were experienced by 39.6% of patients in the pembrolizumab plus chemotherapy group and 26.3% in the placebo plus chemotherapy group and were grade ≥3 in 9.7% and 4.1%, respectively (Table 2).

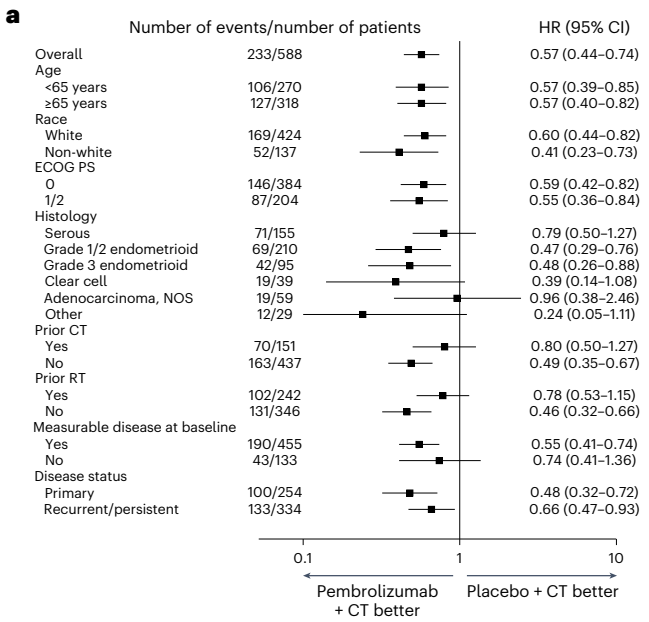
Discussion

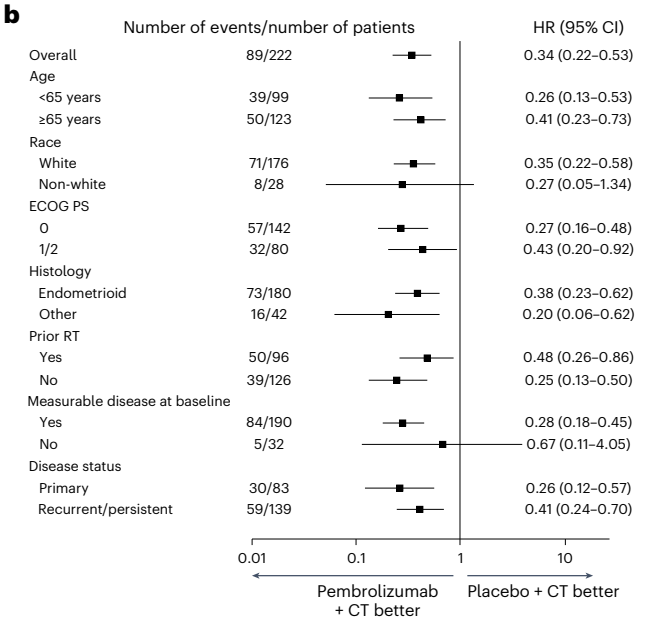
At the IA, the NRG GY018 study met its primary endpoint, with the addition of pembrolizumab (versus placebo) to chemotherapy significantly prolonging investigator-assessed PFS regardless of MMR status[8]. As noted, patients were categorized as pMMR or dMMR somewhat differently in the current analysis (the intention to treat (ITT) was decided using a value assigned at stratification whether from a central or local result) compared with the previous analysis[8]. It is not surprising, however, that outcomes showed limited differences given the high level of concordance between central and local MMR testing. Investigator-assessed PFS HRs were 0.57 in the current analysis and 0.54 in the previous analysis for the pMMR population and 0.34 and 0.30, respectively, for the dMMR population[8].

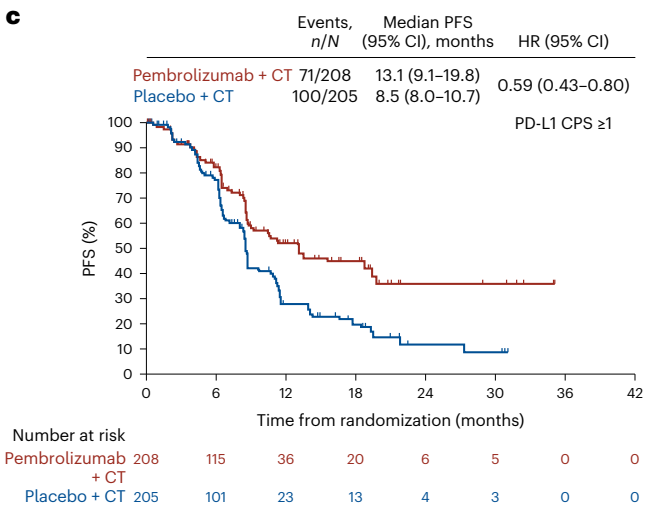
Notably, investigator-assessed PFS favored pembrolizumab plus chemotherapy in all subgroups, including those based on PD-L1 CPS (both populations), methylation (assessed in dMMR population) and histology (both populations). Importantly, in both the pMMR and dMMR populations, results for PFS were consistent in showing clinically meaningful improvement regardless of assessment by investigators or blinded independent central review (BICR) with risk reductions of 43% and 36%, respectively, in the pMMR population and 66% and 55%, respectively, in the dMMR population. Differences between BICR and investigator assessment are expected. Generally, BICR has a more limited and conservative approach to declaring progressive disease by response evaluation criteria in solid tumors (RECIST) v.1.1. This phenomenon, where progressive disease is declared at a later time point by BICR versus investigator, is often observed in clinical studies, whether blinded or open-label, as investigators, unlike BICR, have knowledge of corollary data (including but not limited to signs, symptoms, laboratory abnormalities and procedures such as biopsy with pathological assessment or supplemental imaging). Consistent findings between investigator- and BICR-assessed PFS were also reported in the GOG-3031/RUBY[11] and GOG-3041/DUO-E studies[12].

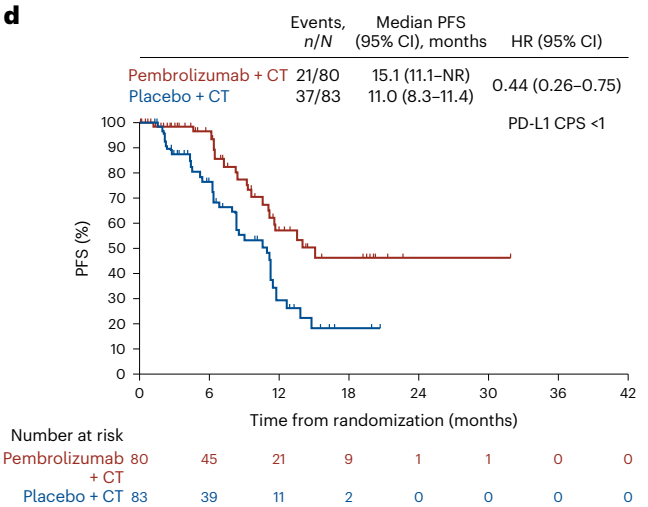
Most patients had PD-L1 CPS ≥ 1 tumor in both the pMMR and dMMR populations. In the pMMR population, addition of pembrolizumab to chemotherapy reduced the risk of investigator-assessed disease progression or death by 41% and 56% in patients with PD-L1

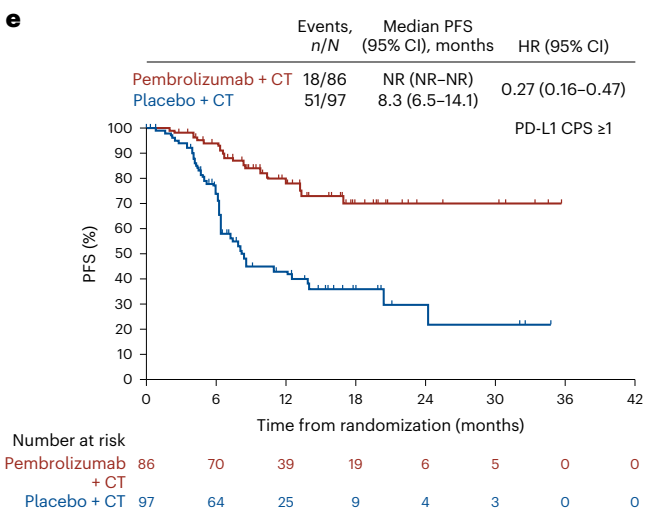
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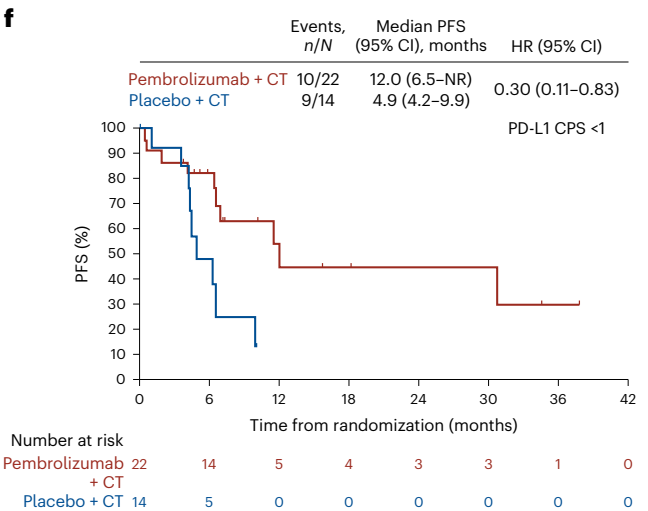


Fig. 4 | Kaplan–Meier-estimated PFS per RECIST v.1.1 as assessed by the investigators in subgroups defined by demographics and baseline clinical characteristics at IA. a,**b**, Kaplan–Meier-estimated PFS per RECIST v.1.1 in subgroups defined by demographics and baseline clinical characteristics in the pMMR population (**a**) and dMMR population (**b**) at IA. **c**–**f**, Results are also shown for the pMMR population with PD-L1 CPS ≥ 1 tumor (**c**), pMMR population with PD-L1 CPS < 1 tumor (**d**), dMMR population with PD-L1 CPS ≥ 1 tumor (**e**) and dMMR population with PD-L1 CPS < 1 tumor (**f**) at IA. Data shown in panels A and B are the HRs (pembrolizumab plus chemotherapy versus placebo plus chemotherapy) and the corresponding 95% CIs. Data cutoff dates: 6 December 2022 (pMMR); 16 December 2022 (dMMR). Subgroup analyses of PFS were based on an unstratified Cox regression model. CPS was defined as the number of PD-L1-positive cells (tumor cells, lymphocytes and macrophages) divided by the total number of tumor cells, multiplied by 100. RT, radiotherapy.

https://doi.org/10.1038/s41591-025-03566-1

Table 2 | Safety at ad hoc analysis (safety population)

AE	Pembrolizumab + chemotherapy (*n*=391)	Placebo + chemotherapy (*n*=388)
Any AE	388 (99.2)	387 (99.7)
Treatment-related AEs	379 (96.9)	373 (96.1)
Most common AEs (≥20% of patients in either treatment group)		
Fatigue	275 (70.3)	248 (63.9)
Anemia	234 (59.8)	220 (56.7)
Alopecia	215 (55.0)	223 (57.5)
Nausea	200 (51.2)	178 (45.9)
Constipation	184 (47.1)	162 (41.8)
Diarrhea	165 (42.2)	138 (35.6)
Peripheral sensory neuropathy	146 (37.3)	158 (40.7)
Decreased white blood cell count	137 (35.0)	134 (34.5)
Peripheral neuropathy	135 (34.5)	116 (29.9)
Decreased platelet count	131 (33.5)	102 (26.3)
Arthralgia	128 (32.7)	140 (36.1)
Decreased neutrophil count	111 (28.4)	114 (29.4)
Dyspnea	97 (24.8)	72 (18.6)
Decreased lymphocyte count	94 (24.0)	75 (19.3)
Hyperglycemia	93 (23.8)	71 (18.3)
Decreased appetite	88 (22.5)	89 (22.9)
Vomiting	83 (21.2)	50 (12.9)
Hypomagnesemia	82 (21.0)	67 (17.3)
Myalgia	79 (20.2)	70 (18.0)
Immune-mediated AEs and infusion reactions	155 (39.6)	102 (26.3)
Infusion reactions	74 (18.9)	72 (18.6)
Hypothyroidism	54 (13.8)	15 (3.9)
Hyperthyroidism	32 (8.2)	10 (2.6)
Severe skin reactions	15 (3.8)	6 (1.5)
Colitis	8 (2.0)	3 (0.8)
Adrenal insufficiency	5 (1.3)	1 (0.3)
Pneumonitis	5 (1.3)	2 (0.5)
Myositis	3 (0.8)	1 (0.3)
Uveitis	3 (0.8)	1 (0.3)
Gastritis	2 (0.5)	0
Hypophysitis	2 (0.5)	0
Myasthenic syndrome	2 (0.5)	0
Nephritis	2 (0.5)	0
Type 1 diabetes mellitus	2 (0.5)	0
Thyroiditis	2 (0.5)	0
Vasculitis	2 (0.5)	1 (0.3)
Encephalitis	1 (0.3)	0
Guillain-Barre syndrome	1 (0.3)	0
Myocarditis	1 (0.3)	0
Pancreatitis	1 (0.3)	0
Sarcoidosis	1 (0.3)	0

All data are *n* (%). Data cutoff date: 18 August 2023.

CPS ≥ 1 and <1 tumor, respectively. Respective risk reductions in the dMMR population were 73% and 70%. In the GOG-3041/DUO-E study, most patients (67.3% overall) had PD-L1-positive tumors (Ventana SP263 IHC assay)[12]. For the overall population, an exploratory analysis was performed showing that the addition of durvalumab to chemotherapy reduced the risk of investigator-assessed disease progression or death by a greater magnitude in patients with PD-L1-positive tumors (HR, 0.63; 95% CI, 0.48–0.83) compared with PD-L1-negative tumors (HR, 0.89; 95% CI, 0.59–1.34)[12]. The discrepancy between NRG GY018 and GOG-3041/DUO-E could be related to the assay used (for example, different antibodies and PD-L1 scoring methods) or the therapy (anti-PD-1 with pembrolizumab versus anti-PD-L1 with durvalumab).

Analysis of OS at IA suggested a directionally favorable benefit with the addition of pembrolizumab versus placebo to chemotherapy in both the pMMR and dMMR populations. This benefit was maintained at 9 months additional follow-up, despite substantial post-study immunotherapy, and improved with sensitivity analysis accounting for the post-study immunotherapy.

As the DCO was in close proximity to the last patient enrolled and not all patients were randomized at this stage, some patients in our study did not have a post-baseline assessment available for response evaluation at the DCOs used for IA. An ad hoc analysis was therefore conducted with approximately 9 additional months of follow-up demonstrating higher response rates, depth of response (including complete response rate) and durability with the addition of pembrolizumab to chemotherapy.

Despite the biostatistical parameters of IA, where the power to detect a difference was modest, especially in the pMMR population, the study met its primary endpoint secondary to robust improvement in PFS when adding pembrolizumab to carboplatin and paclitaxel. This led to the Data Monitoring Committee recommending discontinuation of the study with sponsor unblinding to investigators and patients to their treatment arm so that the vast majority of participants would discontinue placebo. Due to the study being conducted in regions where there is widespread access to immunotherapy, as expected a substantial portion of patients received post-study immunotherapy especially in the control arm. Sensitivity analysis accounting for this discrepancy still suggested a favorable OS benefit with the addition of pembrolizumab. Patients continue to be followed in the study; updated OS results, which were immature at the time of both the IA and ad hoc analyses, will be reported in the future. The study was a collaborative effort led by Cancer Therapy Evaluation Program and NRG Oncology and a registrational trial for the manufacturer of pembrolizumab. Variations in statistical methodology used in this Article compared with the previous publication[8] (for example, the categorization approach for separating patients into pMMR and dMMR populations, and statistical methodology details) were driven by increased harmonization with analytical approaches for other registrational KEYNOTE studies evaluating pembrolizumab that employed ITT principles and did not result in any meaningful differences in numerical values or study conclusions.

No safety concerns were identified with longer follow-up. Results continued to show that rates of chemotherapy-related AEs were not increased with the addition of pembrolizumab. In both treatment groups, rates of immune-mediated AEs and infusion reactions were slightly higher in this ad hoc analysis (39.6% with pembrolizumab plus chemotherapy, 26.3% with placebo plus chemotherapy) than in the primary publication (34.8% and 21.6%, respectively)[8]. Rates of immune-mediated AEs in the pembrolizumab arm were consistent with those previously described for pembrolizumab monotherapy in patients with endometrial cancer[3].

In conclusion, addition of pembrolizumab to first-line chemotherapy and continued as maintenance significantly prolonged investigator-assessed PFS. Secondary analysis showed HRs for OS favoring the pembrolizumab arm, despite substantial rates of post-study immunotherapy, in patients with advanced, metastatic or recurrent endometrial cancer regardless of MMR status. Combining

pembrolizumab with chemotherapy improved investigator-assessed PFS regardless of PD-L1 expression, mechanism of MMR loss or histology, and led to more responses and better durability in both the pMMR and dMMR endometrial cancer populations. PFS results were consistent when assessed by investigators versus BICR. These findings are supportive of the addition of pembrolizumab to chemotherapy as a first-line treatment for patients with advanced, metastatic or recurrent endometrial cancer regardless of MMR status.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41591-025-03566-1.

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Methods

Patients

Eligible patients in the NRG GY018 study (ClinicalTrials.gov NCT03914612) were women ≥18 years old with confirmed advanced (locoregional advanced stage III and IVA), metastatic (stage IVB) or recurrent endometrial cancer. Key inclusion criteria included newly diagnosed stage III or IVA endometrial cancer with measurable disease per RECIST v.1.1 or stage IVB or recurrent endometrial cancer, with or without measurable disease; ECOG PS ≤2; and adequate hematologic, renal and hepatic function. Patients with carcinosarcoma were not eligible for enrollment. Patients who received previous radiation or hormonal therapy were eligible. Prohibited previous treatments included immunotherapy (agents targeting PD-1, PD-L1 or cytotoxic T-lymphocyte antigen 4) and chemotherapy except for adjuvant chemotherapy if the treatment-free interval was ≥12 months. Central MMR IHC testing was required; local testing with central confirmation was also allowed (except for sites in Japan). Further details can be found in the study protocol (https://www.nejm.org/doi/suppl/10.1056/NEJMoa2302312/suppl_file/nejmoa2302312_protocol.pdf).

The study was conducted in accordance with the International Council for Harmonisation Good Clinical Practice guidelines and ethical principles that have their origin in the Declaration of Helsinki. The protocol was approved by an ethics body at each study site (see Supplementary Information for a list of study sites). Patients provided written informed consent. Patients were not compensated for their participation in the study.

Study design and treatment

In this phase 3, international, double-blind study, patients were randomized (1:1) to receive either pembrolizumab plus paclitaxel and carboplatin or placebo plus paclitaxel and carboplatin (six cycles) followed by pembrolizumab or placebo maintenance (up to 14 cycles). Randomization was stratified according to MMR status (dMMR versus pMMR), ECOG PS (0/1 versus 2) and receipt of previous adjuvant chemotherapy (yes versus no). All treatments were administered intravenously. Pembrolizumab 200 mg (or placebo), paclitaxel 175 mg m^{-2}, and carboplatin area under the curve 5 mg ml^{-1} min^{-1} were administered on day 1 of each 3-week cycle during the first six cycles. Pembrolizumab 400 mg or placebo was administered once every 6 weeks during maintenance.

The study was unblinded after meeting its primary endpoint of investigator-assessed PFS at IA (DCOs of 6 December 2022 and 16 December 2022 for pMMR and dMMR populations, respectively). Investigators were informed of patients' assigned treatment and patients were made aware of the study outcome and their treatment assignment. Consequently, nearly all patients (>99%) in the placebo plus chemotherapy group discontinued study treatment by the time of the ad hoc analysis reported herein (DCO, 18 August 2023), with some participants subsequently receiving immunotherapy outside the study. There were no clinically important protocol deviations; the approved protocol amendments are reflected in this Article and detailed in the study protocol.

Endpoints

Results for the primary endpoint, PFS per RECIST v.1.1 as assessed by the investigator in the pMMR and dMMR populations, were previously reported[8]. Secondary endpoints reported herein include concordance between local and central MMR IHC testing; OS; ORR, best overall response and DOR per RECIST v.1.1 as assessed by the investigator; and safety. Exploratory endpoints presented include the association between PD-L1 CPS with MMR status, subgroup analyses of investigator-assessed PFS per RECIST v.1.1 and PFS per RECIST v.1.1 as assessed by BICR. Secondary and exploratory endpoints were assessed separately in the pMMR and dMMR populations except for safety (combined population) and subgroup analyses of investigator-assessed PFS based on methylation status (dMMR population only).

Assessments

Radiographic tumor measurements were performed within 28 days before treatment (baseline), every 9 weeks for the first 9 months, and every 12 weeks thereafter until disease progression or initiation of subsequent anticancer therapy (see 'Assessments' in Supplementary Information for further details). Formalin-fixed, paraffin-embedded unstained sections were provided for central MMR IHC testing; local MMR IHC testing was performed per institutional practice. PD-L1 expression in tumor samples was determined centrally using PD-L1 IHC 22C3 pharmDx (Agilent Technologies). Tumors with PD-L1 CPS ≥1 were considered PD-L1 positive. AEs that occurred from randomization 30 days after the last dose of study drug were graded per National Cancer Institute Common Terminology Criteria for Adverse Events v.5.0.

Statistical analyses

The expected number of PFS events for the final analysis was 394 (~196 at IA) in the pMMR population and 168 (~84 at IA) in the dMMR population. The initial alpha level for PFS hypothesis testing was set at 0.0125 1-sided and could be transferred to the other population upon achieving statistical significance. The expected number of deaths for the final analysis was 364 in the pMMR population and 150 in the dMMR population. Statistical analyses for secondary and exploratory endpoints are descriptive in nature; *P* values are nominal. Concordance between local and central MMR IHC testing was characterized using a Cohen kappa coefficient. PFS, OS and DOR were analyzed using the Kaplan–Meier method for censored data. For both PFS and OS, data were censored at the last contact date if the event had not occurred. HRs and 95% CIs for PFS and OS were based on a Cox regression model with the Efron method of tie handling with treatment as a covariate stratified by prior chemotherapy. One-sided *P* values were based on the log-rank test stratified by prior chemotherapy. Subgroup analyses of PFS were based on an unstratified Cox regression model. Gender analyses were not carried out as all eligible patients were female. The 95% CIs for objective responses were based on the binomial exact method; between-group differences and 95% CIs were based on the Miettinen and Nurminen method stratified by prior chemotherapy.

The primary DCO for the IA were 6 December 2022 for the pMMR population and 16 December 16 for the dMMR population. Efficacy analyses at IA (PFS per investigator assessment including in subgroups, PFS by BICR and OS) were conducted in the ITT population. Because some patients had no post-baseline assessment available for response evaluation at the December 2022 DCOs, an ad hoc analysis of ORR, best overall response and DOR was performed using a later DCO on 18 August 2023. Patients had to have measurable disease at baseline to be included in the analyses of objective responses. OS in the ITT population, along with sensitivity analysis for post-study subsequent anticancer therapy, and safety in all randomized and treated patients were also assessed as part of the ad hoc analysis. Statistical methods for the confounding effect of subsequent anticancer therapy on OS are described in Supplementary Information.

The current analyses and those described previously[8] used different methods of categorizing patients into either the pMMR or dMMR population. In the previous analyses, central MMR status was used, if available, with local MMR status being used in the absence of central results. Patients with indeterminate central MMR status were excluded as per protocol. For all of the current analyses presented here, patients were categorized based on the MMR stratification status used for randomization, which could have been based on either central or local MMR results. At the end of randomization, 819 patients had been randomized in the study. Nine patients in the pMMR population were randomized after the 6 December 2022 DCO and therefore were not included in both the current and previous PFS and OS analyses using the IA DCOs, but are included in the ad hoc analyses (18 August 2023 DCO).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc. (MSD), is committed to providing qualified scientific researchers access to anonymized data and clinical study reports from the company's clinical trials for the purpose of conducting legitimate scientific research. MSD is also obligated to protect the rights and privacy of trial participants and, as such, has a procedure in place for evaluating and fulfilling requests for sharing company clinical trial data with qualified external scientific researchers. The MSD data sharing website (available at https://externaldatasharing-msd.com/) outlines the process and requirements for submitting a data request. Applications will be promptly assessed for completeness and policy compliance. Feasible requests will be reviewed by a committee of MSD subject matter experts to assess the scientific validity of the request and the qualifications of the requestors. In line with data privacy legislation, submitters of approved requests must enter into a standard data sharing agreement with MSD before data access is granted. Data will be made available for request after product approval in the United States and European Union or after product development is discontinued. There are circumstances that may prevent MSD from sharing requested data, including country- or region-specific regulations. If the request is declined, it will be communicated to the investigator. Access to genetic or exploratory biomarker data requires a detailed, hypothesis-driven statistical analysis plan that is collaboratively developed by the requestor and MSD subject matter experts; after approval of the statistical analysis plan and execution of a data sharing agreement, MSD will either perform the proposed analyses and share the results with the requestor or will construct biomarker covariates and add them to a file with clinical data that is uploaded to an analysis portal so that the requestor can perform the proposed analyses.

Acknowledgements

This work was supported by grants U10CA180868 and U10CA180822 from the National Cancer Institute (NCI). Funding and support were received from Merck & Co., Inc., Rahway, NJ, USA through a Cooperative Research and Developmental Agreement with NCI. Merck & Co., Inc. also provided supplemental funding to NRG Oncology for this study. Support was provided in part by the National Institutes of Health (NIH)/NCI Cancer Center Support Grant P30CA008748 (to C.A. and R.E.O.C.) and NIH-UG1 CA233330 (to C.A.L.). Funding for this research was provided by Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc. (MSD). The protocol was developed by NRG Oncology. MSD had a role in the current data analysis, the decision to publish and paper review. We thank the patients and their families and caregivers for participating in this study, along with all investigators and site personnel. Medical writing assistance was provided by M. S. McNamara of ICON plc (Blue Bell, PA, USA). This assistance was funded by MSD. The following NRG Oncology Group member institutions participated in the primary treatment studies: CWRU Case Comprehensive Cancer Center, University of Rochester, Pacific Cancer Research Consortium, University of Oklahoma Health Sciences Center, Thomas Jefferson University Hospital, Georgia NCI Community Oncology Research Program, Rutgers Cancer Institute of New Jersey, Women and Infants Hospital, Kaiser Permanente NCI Community Oncology Research Program, University of Alabama at Birmingham/Deep South Research Consortium, Odette Cancer Centre – Sunnybrook Health Sciences Centre, Northwestern University, Roswell Park Comprehensive Cancer Center, Indiana University/Melvin and Bren Simon Cancer Center, Ohio State University Comprehensive Cancer Center, Memorial Sloan-Kettering Cancer Center, Michigan Cancer Research Consortium NCORP, University of Iowa/Holden Comprehensive Cancer Center, Washington University – Siteman Cancer Center, Northwell Health NCORP, UC San Diego Moores Cancer Center, CommonSpirit Health Research Institute, NorthShore University HealthSystem at Evanston Hospital, University of Michigan Comprehensive Cancer Center, UPMC Hillman Cancer Center, ProMedica Flower Hospital, Sanford NCI Community Oncology Research Program of the North Central Plains, West Virginia University Charleston Division, Nebraska Methodist Hospital, Mayo Clinic, University of Nebraska Medical Center, University of Cincinnati Cancer Center – UC Medical Center, Houston Methodist Hospital, Cancer Research Consortium of West Michigan NCORP, University of Kansas Cancer Center – MCA Rural MU NCORP, Greater Baltimore Medical Center, Carolinas Medical Center/Levine Cancer Institute, New Mexico Minority Underserved NCORP, Avera Cancer Institute, University of Virginia Cancer Center, Legacy Good Samaritan Hospital and Medical Center, Saitama Medical University International Medical Center, Columbus NCI Community Oncology Research Program, JHU Sidney Kimmel Comprehensive Cancer Center, Puerto Rico Minority Underserved NCORP, West Virginia University Healthcare, Mount Sinai Hospital, UC Irvine Health/Chao Family Comprehensive Cancer Center, Lahey Hospital and Medical Center, Montana Cancer Consortium NCORP, Allegheny General Hospital, VCU Massey Cancer Center Minority Underserved NCORP, CHUM – Centre hospitalier de l'Université de Montréal, University Health Network – Princess Margaret Hospital, UCLA/Jonsson Comprehensive Cancer Center, Gulf South Minority Underserved NCORP, Heartland Cancer Research NCORP, Yale University – Yale Cancer Center LAPS, Montefiore Minority Underserved NCORP, Wake Forest University Health Sciences, New York-Presbyterian/Brooklyn Methodist Hospital, Sutter Cancer Research Consortium, UNC Lineberger Comprehensive Cancer Center LAPS, Cancer Research for the Ozarks NCORP, WellSpan Health – York Hospital, Southeast Clinical Oncology Research Consortium NCORP, Baylor College of Medicine/Dan L. 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Author contributions

The study concept and design was provided by R.N.E., M.W.S., C.A., A.N.F. and M.A.P. The provision of materials or patients is accredited to R.N.E., C.A., L.B., R.G.M., J.M.H., F.B.M., R.S.M., M.S.S., G.H.C., E.G., E.L., J.K., C.A.L., L.T.G., E.M.H., S.B.L, L.M.L., F.B., R.E.O.C.,

T.A.B., E.K.H., P.H.T., V.S.J., S.W., A.N.F. and M.A.P. Data acquisition was provided by R.N.E., M.W.S., C.A., R.G.M., J.M.H., F.B.M., R.S.M., M.S.S., E.G., J.K., C.A.L., E.M.H., S.B.L., R.E.O.C., A.N.F. and M.A.P. Data analysis, interpretation and verification are accredited to R.N.E., C.A., R.S.M., M.S.S., G.H.C., E.G., R.E.O.C. and A.N.F. We thank R.N.E., C.A., R.S.M., E.G., L.M.L., P.H.T. and A.N.F. for writing the paper. All authors contributed to the critical review of the paper and the final approval.

Competing interests

Funding for this research was provided by Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc. (MSD). The protocol was developed by NRG Oncology. MSD had a role in the current data analysis, the decision to publish and manuscript review. R.N.E. received institution research support from AstraZeneca, Clovis Oncology, Eisai, MSD and Novocure; consulting and advisory fees from AstraZeneca, Cardiff Oncology, Clovis Oncology, Eisai, Elevar Therapeutics, GSK/Tesaro, ImmunoGen, Inc., Seagen, Mersana, Myriad, Daichi Sankyo, PMV, Gilead, Onconova, Elevar Therapeutics and Regeneron along with personal payments for lectures and presentations from AstraZeneca, GSK, Immunogen, Merck, Medscape, Curio Biosciences, Great Debates and Updates. Additionally, R.N.E. serves in a leadership role as GOG Associate Clinical Trial Advisor as well as Scientific and Medical Advisor to the Clearity Foundation. M.W.S. reports receiving support from his employer, the Roswell Park Comprehensive Cancer Center with support from the National Cancer Institute as the government agency that funds the research. M.W.S. also acknowledges receiving support to attend meetings and travel R.G.M. received consulting fees from Fujirebio Diagnostics, Inc. F.B.M. received institutional support for clinical trials from GSK, Clovis, Immunogen, AstraZeneca, Seagen, Duality Bio, Acrivon, Allarity and Adaptimune, as well as consulting fees from the advisory boards of AstraZeneca, GSK, Daiichi Sankyo, BioNTech, Aadi. Consulting fees were also received from GSK and Seagen and an additional speaker fee from MSD. F.B.M. paritcipated on advisory boards for AstraZeneca, GSK, Daiichi Sankyo, BioNTech and Aadi and also reports serving on the Board of Directors for the Society of Gynecologic Oncology (unpaid) as well as on the Board of Directors for the Rivkin Foundation (uncompensated). R.S.M. serves on the NRG Board of Directors for which his Institution receives compensation. M.S.S. receives consulting fees and honoraria from GSK, AstraZeneca, Merck and Immunogen and also serves as a board member for Unite for HER, a non-profit organization. E.G. served on an advisory board for Merck. E.L. received a grant from the Rhode Island Foundation. J.K. reports that US$300 was paid to their institution, Kaiser Permanente, in March 2023 for a lecture to the Association of Northern California Oncologists (ANCO) as well as receiving hotel expenses for this meeting. J.K. participated on an advisory board for a mirvetuximab study and approximately US$600 was paid to the institution in May 2022. C.A.L. received research funding for their institution and personal consulting fees, honorarium and international travel expenses from Merck. L.T.G. was paid consulting fees from Merck and GSK and also paid honoraria by AstraZeneca as a speaker. E.M.H. received consulting fees from AbbVie/Immunogen. L.M.L. received consulting fees for her participation on GlaxoSmithKline's advisory board. Research grants were paid to the institution of F.B. from Merck, Eisai, ImmunoGen, Clovis, Beigene, Natera, Tempus and AstraZeneca while personal fees were also obtained from UptoDate and advisory boards (personal fees) were received from Agenus, Merck, Clovis, Immunogen, Eisai, AstraZeneca, GlaxoSmithKline, Myriad, BioNTech, Daiichi Sankyo, EMD Serono. Personal fees were also received by F.B. for CME lectures from Clinical Educational Concepts, Clinical Care Options, Medscape/WebMD, Med Learning, I3Health, CMR Institute, Global Learning Initiative/Prova, OncLive, Targeted Oncology, Research to Practice, GOG Foundation. GlaxoSmithKline and BioNTech provided support for F.B. to attend meetings and travel. F.B. holds uncompensated leadership roles on the Society of Gynecologic Oncology Board Member as a Co-Chair; NRG Oncology Developmental Therapeutic as a committee member and IGCS Education 360 as a Co-Chair. R.E.O.C. had support for this paper from NCI/NIH (P30CA008748). Grants were paid to R.E.O.C.'s institution from Bayer/Celgene/Juno, Arsenalbio, Tesaro/GSK, Merck, Ludwig Cancer Institute, AbbVie/StemCentrx, Regeneron, TCR2 Therapeutics, Atara Biotherapeutics, Inc., Marker Therapeutics, Syndax Pharmaceuticals, Genmab/Seagen Therapeutics, Genentech, Alkermes/Mural Oncology, Kite Pharma, Acrivon, OnCusp Therapeutics, and Gynecologic Oncology Foundation and the Lyell Immunopharma. R.E.O.C. also received payment and honoraria (one-off lectures) from GSK, Curio/Onclive/PER/MJH/Aptitude Health; SITC; Gynecologic Oncology Canada along with support for attending meetings from Hitech Health; Gathering Around Cancer, Ireland; GOG Foundation and SGO. In addition, R.E.O.C. served in an unpaid capacity for the Steering Committee for AstraZeneca (DUO-0), GSK (Moonstone, Prima) and Acrivon, as well as an unpaid advisor for Carina Biotech and Link Therapeutics. Furthermore, R.E.O.C. was an advisor to OnCusp Therapeutics and served on advisory boards for Tesaro/GSK, Regeneron Advisory, Seattle Genetics/Seagen, Immunogen Board Bayer, R Pharm, Fresenius Kabi, Miltenyi, 2seventy bio, Bayer and Loxo. R.E.O.C. is Vice-Chair for CPC and SGO and Chair of the DT Committee for NRG Oncology. The institution of P.H.T. received grants from Merck and GlaxoSmithKline. P.H.T. has stock in Immunon and was personally compensated for serving on advisory boards for Iovance, AstraZeneca, Verastem, GlaxoSmithKline, Seagen, Immunon, Immunogen, Mersana, Novocure, Zentalis, Caris, Merck and Caris. S.W. received personal payments and honoraria from Merck, GSK, AstraZeneca and Eisai. A.N.F. received honararia for speaking at the GOG Foundation, Endometrial Cancer Symposium and has also participated on the GSK advisory board while also serving as President of the Society of Gynecologic Oncology (unpaid position). M.A.P. received personal consulting fees from GSK, Tesaro, Merck, AstraZeneca, Immunogen, Seagen, Clovis Oncology and Kyropharma. M.A.P. also served in leadership roles for the GOG Foundation, SGO and FWC (uncompensated) alongside his role with NRG Oncology (compensated). C.A. received clinical trial funding to their institution (MSK) from AbbVie – MSK PI, GOG3005, Artios Pharma – International Coordinating Investigator & MSK PI, Artist; AstraZeneca – MSK PI, SOLO1/GOG 3004; National Coordinating Investigator & MSK PI, D081RC00001; ENGOT – ov46; AGO-OVAR 23; GOG-3025; Clovis – MSK PI, ARIEL 2 & 3; Genentech/Roche – MSK PI, GOG3015 (imAGYN050). C.A. participated in a virtual advisory board for Blueprint Medicine on 30 June 2021 (no consulting fee received) and also attended a virtual advisory board for Merck for Global Cervical and Ovarian Cancer on 10 July 2023. C.A. has ongoing commitments with AstraZeneca – AZ eVOLVE DMC 26 April 2023; AstraZeneca – Future Clinical Strategies for Endometrial Cancer Virtual Advisory Board 13 March 204 and Verastem – RAMP 301 Steering Committee, 6 March 2024. C.A. participated on a data safety monitoring board and advisory board for Blueprint Medicine – Virtual Advisory Board on 30 June 2021 (no consulting fee received); Merck – Global Cervical and Ovarian Cancer Virtual Advisory Board 10 July 2023; AstraZeneca – AZ eVOLVE DMC 26 April 2023 (ongoing); AstraZeneca – Future Clinical Strategies for Endometrial Cancer Virtual Advisory Board 13 March 2024; Verastem, RAMP 301 Steering Committee, 6 March 2024 (ongoing). C.A. serves on the Board of Directors for both the GOG Foundation and NRG Oncology (unpaid). The other authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41591-025-03566-1.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41591-025-03566-1.

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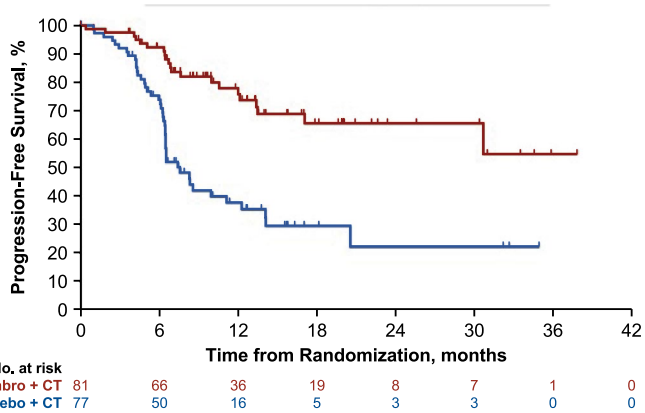
Peer review information *Nature Medicine* thanks Toon Van Gorp and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Ulrike Harjes, in collaboration with the *Nature Medicine* team.

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https://doi.org/10.1038/s41591-025-03566-1

A)

	Events, n/N	Median PFS (95% CI), mo	HR (95% CI)[a]
Pembro + CT	21/81	NR (30.7–NR)	0.30 (0.18–0.50)
Placebo + CT	44/77	7.4 (6.4–11.1)	

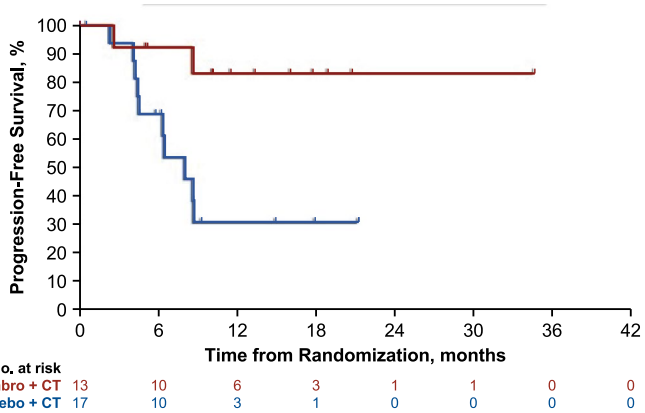


No. at risk

Pembro + CT	81	66	36	19	8	7	1	0
Placebo + CT	77	50	16	5	3	3	0	0

B)

	Events, n/N	Median PFS (95% CI), mo	HR (95% CI)[a]
Pembro + CT	2/13	NR (8.6–NR)	0.17 (0.04–0.79)
Placebo + CT	10/17	8.0 (4.4–NR)	



No. at risk

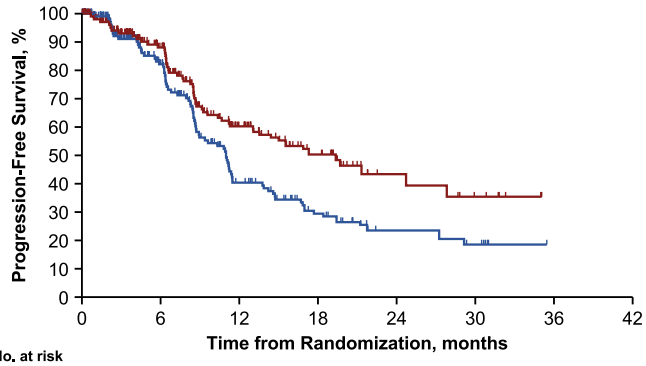
Pembro + CT	13	10	6	3	1	1	0	0
Placebo + CT	17	10	3	1	0	0	0	0

Extended Data Fig. 1 | Kaplan-Meier–estimated PFS per RECIST v1.1 as assessed by the investigators in the dMMR population (A) with methylation and (B) with no methylation at interim analysis. Data cutoff date: December 16, 2022. [a]Based on a Cox regression model with the Efron method of tie handling with treatment as a covariate. CT, chemotherapy; dMMR, mismatch repair-deficient; HR, hazard ratio; NR, not reached; pembro, pembrolizumab; PFS, progression-free survival; pMMR, mismatch repair-proficient; RECIST, Response Evaluation Criteria in Solid Tumors.

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A)

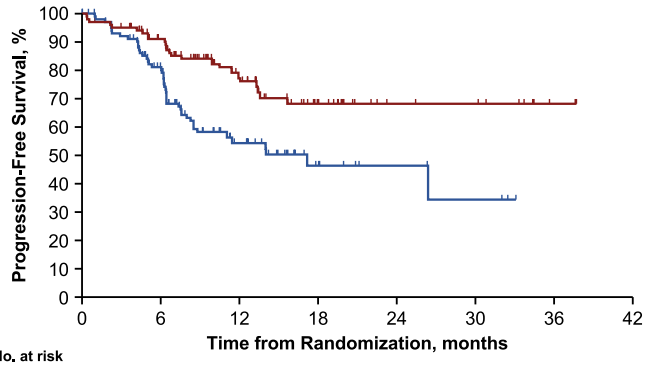
	Events, n/N	Median PFS (95% CI), mo	HR (95% CI)[a], *P*-value[b]
Pembro + CT	85/294	19.5 (13.1–28.0)	**0.64 (0.49–0.85)**
Placebo + CT	122/294	11.0 (9.0–11.5)	***P* = 0.0008**



No. at risk								
Pembro + CT	294	167	64	34	11	7	0	0
Placebo + CT	294	155	48	23	9	6	0	0

B)

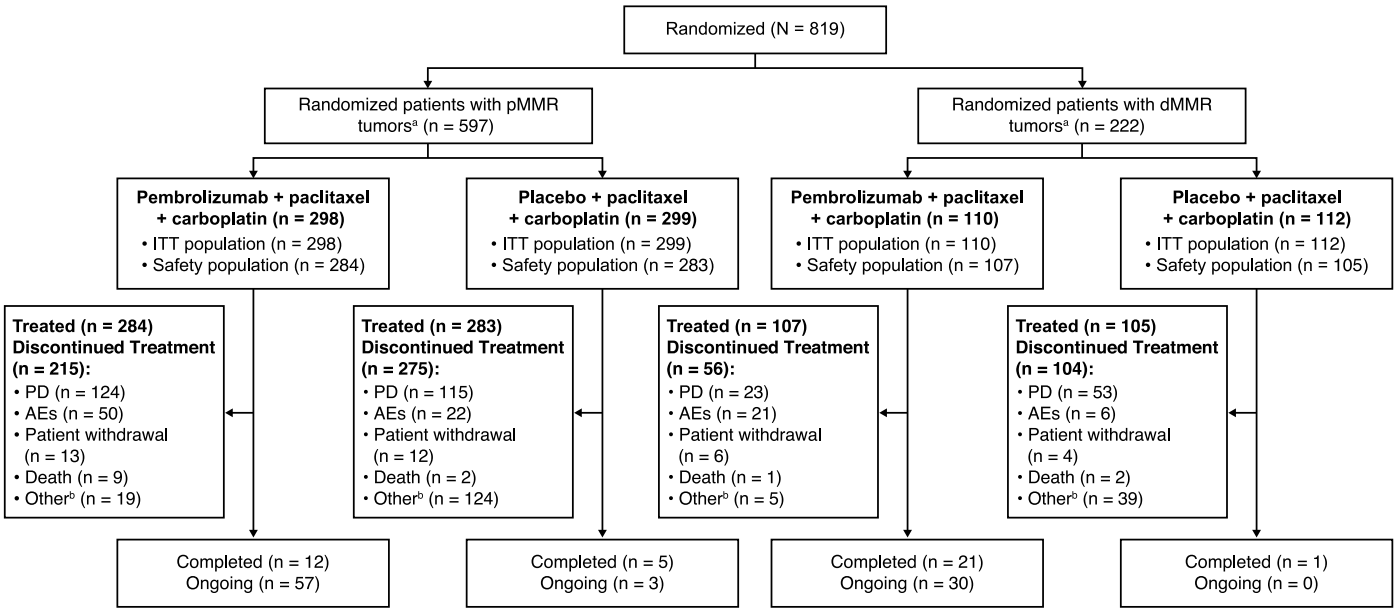
	Events, n/N	Median PFS (95% CI), mo	HR (95% CI)[a], *P*-value[b]
Pembro + CT	25/110	NR (NR–NR)	**0.45 (0.27–0.73)**
Placebo + CT	45/112	14.1 (8.5–NR)	***P* = 0.0005**



No. at risk								
Pembro + CT	110	85	48	26	10	9	2	0
Placebo + CT	112	75	31	10	5	3	0	0

Extended Data Fig. 2 | Kaplan-Meier–estimated PFS per RECIST v1.1 as assessed by BICR in the (A) pMMR and (B) dMMR populations at interim analysis. Data cutoff dates: December 6, 2022 (pMMR); December 16, 2022 (dMMR). [a]Based on a Cox regression model with the Efron method of tie handling with treatment as a covariate stratified by prior chemotherapy. [b]One-sided *P* value based on log-rank test stratified by prior chemotherapy. CT, chemotherapy; dMMR, mismatch repair-deficient; HR, hazard ratio; NR, not reached; pembro, pembrolizumab; PFS, progression-free survival; pMMR, mismatch repair-proficient; RECIST, Response Evaluation Criteria in Solid Tumors.

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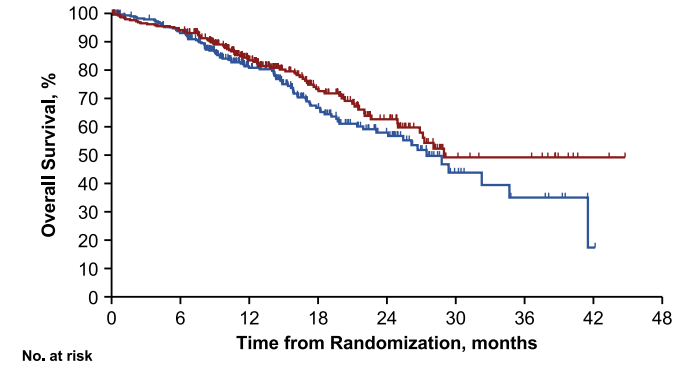


Extended Data Fig. 3 | Patient disposition in the pMMR and dMMR populations at ad hoc analysis. Data cutoff date: August 18, 2023. [a]By central or local immunohistochemistry. [b]Includes alternative therapy (in absence of progression), off treatment due to other complicating disease, symptomatic deterioration, and other (not otherwise specified). AE, adverse event; dMMR, mismatch repair-deficient; ITT, intention to treat; PD, progressive disease; pMMR, mismatch repair-proficient.

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A)

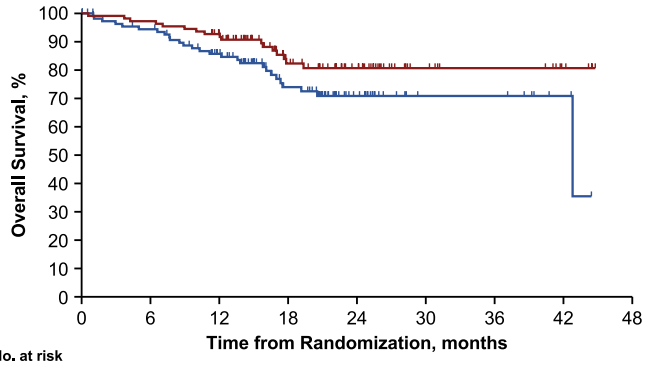
	Events, n/N	Follow-up Duration[a], median (range), mo	Median OS (95% CI), mo	HR (95% CI)[b], *P*-value[c]
Pembro + CT	77/298	15.7 (0.5–45.4)	28.9 (26.8–NR)	**0.80 (0.59–1.08)**
Placebo + CT	92/299	15.0 (0.9–45.6)	28.7 (24.0–34.6)	***P* = 0.0683**



No. at risk									
Pembro + CT	298	271	169	94	49	14	11	3	0
Placebo + CT	299	268	159	89	47	13	7	2	0

B)

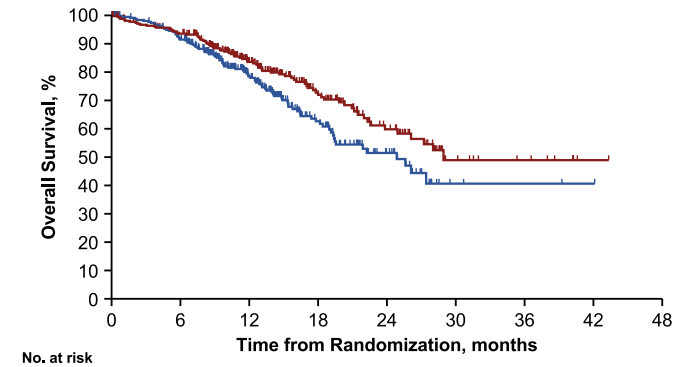
	Events, n/N	Follow-up Duration[a], median (range), mo	Median OS (95% CI), mo	HR (95% CI)[b], *P*-value[c]
Pembro + CT	17/110	19.3 (0.6–47.4)	NR (NR–NR)	**0.57 (0.31–1.04)**
Placebo + CT	27/112	19.0 (1.0–44.8)	42.7 (42.7–NR)	***P* = 0.0323**



No. at risk									
Pembro + CT	110	106	93	53	35	14	10	6	0
Placebo + CT	112	100	82	50	24	8	8	4	0

C)

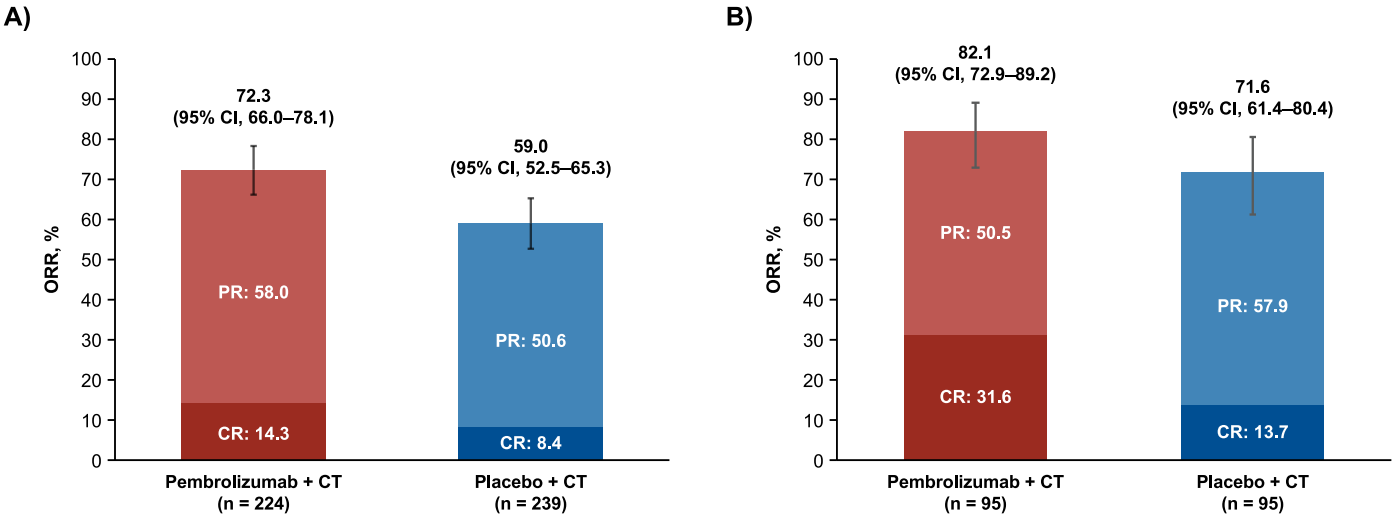
	Events, n/N	Median OS (95% CI), mo	HR (95% CI)[b]
Pembro + CT	77/298	28.9 (24.9–NR)	**0.70 (0.50–0.98)**
Placebo + CT	92/299	24.8 (19.3–NR)	



No. at risk									
Pembro + CT	298	269	166	87	43	12	7	2	0
Placebo + CT	299	262	148	68	28	3	2	1	0

Extended Data Fig. 4 | Kaplan-Meier–estimated OS in the (A) pMMR and (B) dMMR populations and (C) sensitivity analysis in the pMMR population at ad hoc analysis. OS data were immature (46.4% information fraction for pMMR population [169/364 events needed for final analysis had occurred] and 29.3% information fraction for dMMR population [44/150 events needed for final analysis had occurred]). Data cutoff date: August 18, 2023. [a]Follow-up duration is the time from randomization to the date of death or the database cutoff date if the participant is still alive; follow-up duration was not calculated for the sensitivity analysis (panel C) owing to the nature of the bootstrap analysis. [b]Based on a Cox regression model with the Efron method of tie handling with treatment as a covariate stratified by prior chemotherapy. [c]One-sided *P* value based on log-rank test stratified by prior chemotherapy. CT, chemotherapy; dMMR, mismatch repair-deficient; HR, hazard ratio; NR, not reached; OS, overall survival; pembro, pembrolizumab; pMMR, mismatch repair-proficient.

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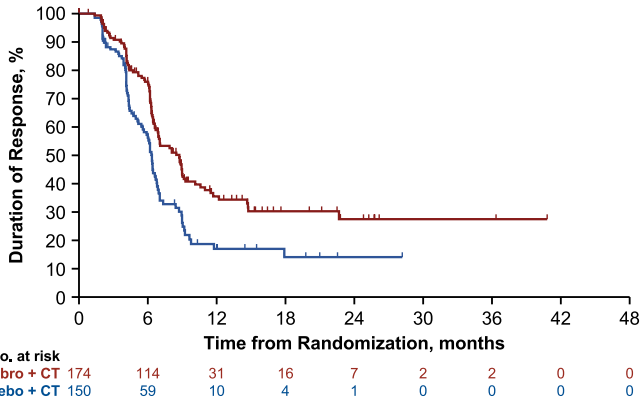


Extended Data Fig. 5 | Investigator-assessed response rates per RECIST v1.1 at ad hoc analysis in the (A) pMMR and (B) dMMR populations. Analysis included patients in the intention-to-treat population with measurable disease at baseline. Data cutoff date: August 18, 2023. Data shown are the mean ORR and 95% CIs based on the binomial exact method. CR, complete response; CT, chemotherapy; dMMR, mismatch repair-deficient; ORR, objective response rate; pMMR, mismatch repair-proficient; PR, partial response; RECIST, Response Evaluation Criteria in Solid Tumors.

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A)

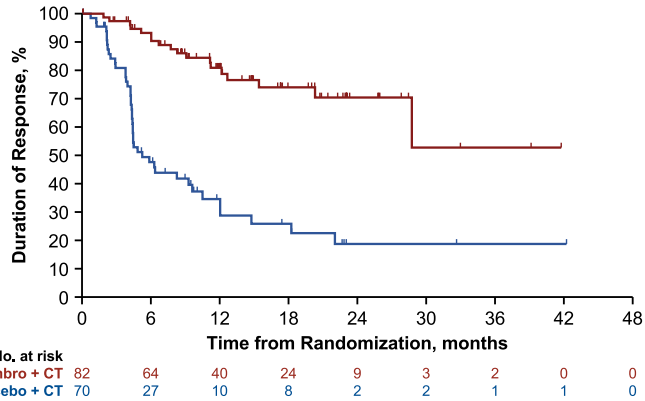
	Median (range) DOR, mo
Pembro + CT	8.1 (0.0+ to 40.9+)
Placebo + CT	6.4 (0.0+ to 28.3+)



No. at risk									
Pembro + CT	174	114	31	16	7	2	2	0	0
Placebo + CT	150	59	10	4	1	0	0	0	0

B)

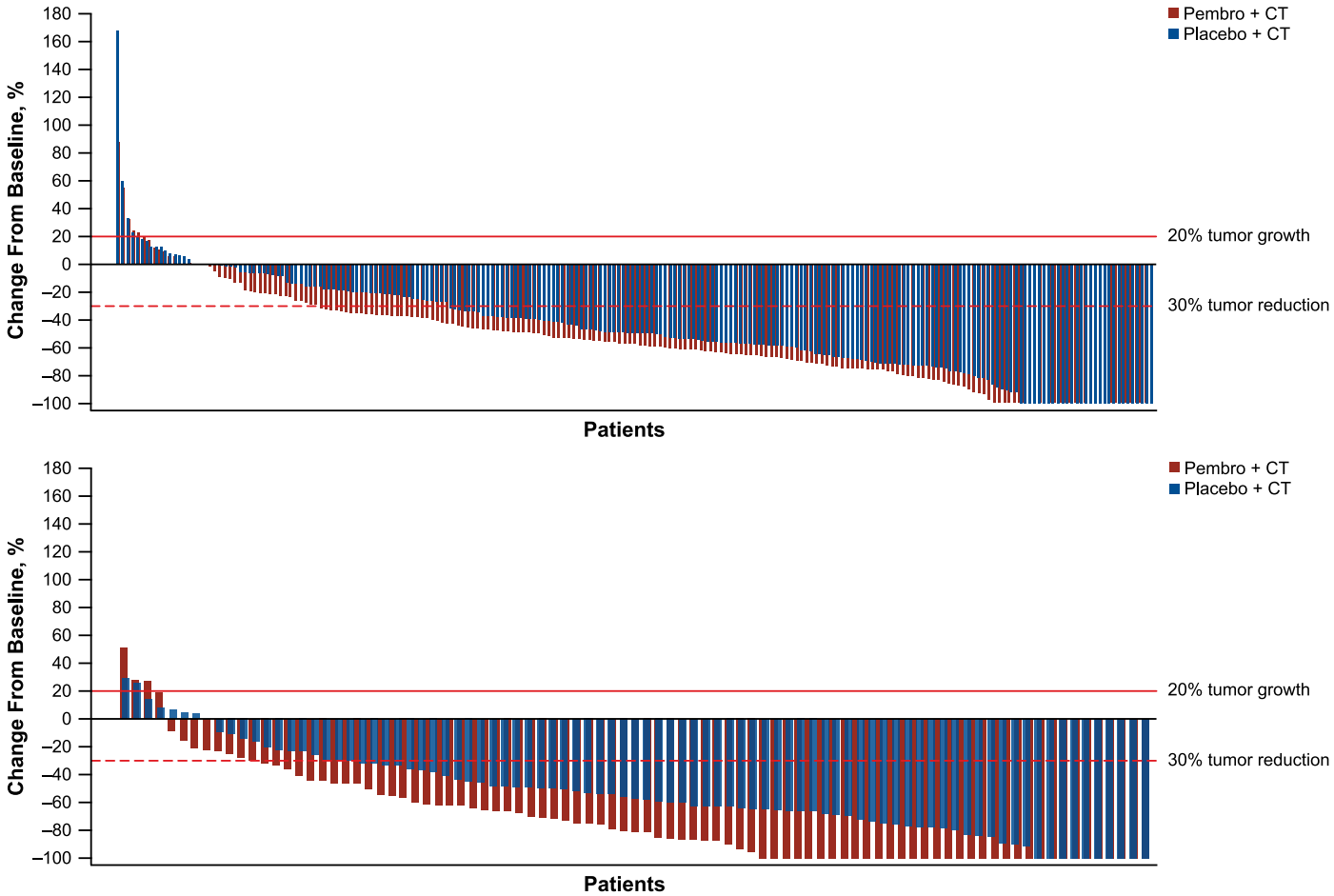
	Median (range) DOR, mo
Pembro + CT	NR (0.0+ to 41.8+)
Placebo + CT	4.8 (0.0+ to 42.2+)



No. at risk									
Pembro + CT	82	64	40	24	9	3	2	0	0
Placebo + CT	70	27	10	8	2	2	1	1	0

Extended Data Fig. 6 | Kaplan-Meier–estimated DOR per RECIST v1.1 as assessed by the investigators in the (A) pMMR and (B) dMMR populations at ad hoc analysis. Data cutoff date: August 18, 2023. +, no progressive disease at last disease assessment; CT, chemotherapy; dMMR, mismatch repair-deficient; DOR, duration of response; NR, not reached; pembro, pembrolizumab; pMMR, mismatch repair-proficient; RECIST, Response Evaluation Criteria in Solid Tumors.

https://doi.org/10.1038/s41591-025-03566-1



Extended Data Fig. 7 | Investigator-assessed best percentage change from baseline in target lesions per RECIST v1.1 in the (A) pMMR and (B) dMMR populations at ad hoc analysis. Data cutoff date: August 18, 2023.

CT, chemotherapy; dMMR, mismatch repair-deficient; pembro, pembrolizumab; pMMR, mismatch repair-proficient; RECIST, Response Evaluation Criteria in Solid Tumors.

https://doi.org/10.1038/s41591-025-03566-1

Extended Data Table 1 | Demographics and Baseline Clinical Characteristics at Ad Hoc Analysis (ITT Population)

	pMMR Population			dMMR Population		
	Pembrolizumab + Chemotherapy (n = 298)	Placebo + Chemotherapy (n = 299)	All Patients (n = 597)	Pembrolizumab + Chemotherapy (n = 110)	Placebo + Chemotherapy (n = 112)	All Patients (n = 222)
Age, median (range), y	66.2 (31.0–94.0)	66.0 (29.0–91.0)	66.1 (29.0–94.0)	67.2 (39.0–82.0)	66.0 (37.0–86.0)	66.1 (37.0–86.0)
Race, n (%)						
White	216 (72.5)	215 (71.9)	431 (72.2)	91 (82.7)	85 (75.9)	176 (79.3)
Black or African American	46 (15.4)	51 (17.1)	97 (16.2)	10 (9.1)	9 (8.0)	19 (8.6)
Asian	17 (5.7)	15 (5.0)	32 (5.4)	3 (2.7)	4 (3.6)	7 (3.2)
American Indian or Alaska Native	2 (0.7)	2 (0.7)	4 (0.7)	0 (0)	2 (1.8)	2 (0.9)
Native Hawaiian or other Pacific Islander	1 (0.3)	3 (1.0)	4 (0.7)	0 (0)	0 (0)	0 (0)
Multiracial	1 (0.3)	1 (0.3)	2 (0.3)	0 (0)	0 (0)	0 (0)
Missing data	15 (5.0)	12 (4.0)	27 (4.5)	6 (5.5)	12 (10.7)	18 (8.1)
Hispanic ethnic group, n (%)						
No	267 (89.6)	278 (93.0)	545 (91.3)	104 (94.5)	97 (86.6)	201 (90.5)
Yes	23 (7.7)	15 (5.0)	38 (6.4)	4 (3.6)	7 (6.3)	11 (5.0)
Unknown	4 (1.3)	3 (1.0)	7 (1.2)	1 (0.9)	4 (3.6)	5 (2.3)
Not reported	4 (1.3)	3 (1.0)	7 (1.2)	1 (0.9)	4 (3.6)	5 (2.3)
ECOG PS, n (%)						
0	200 (67.1)	201 (67.2)	401 (67.2)	70 (63.6)	72 (64.3)	142 (64.0)
1	89 (29.9)	89 (29.8)	178 (29.8)	39 (35.5)	35 (31.3)	74 (33.3)
2	9 (3.0)	9 (3.0)	18 (3.0)	1 (0.9)	5 (4.5)	6 (2.7)
PD-L1 CPS,[a] n	291	294	585	109	111	220
≥1, n (%)	209 (71.8)	209 (71.1)	418 (71.5)	87 (79.8)	97 (87.4)	184 (83.6)
<1, n (%)	82 (28.2)	85 (28.9)	167 (28.5)	22 (20.2)	14 (12.6)	36 (16.4)
Methylation, n	–	–	–	94	94	188
Yes, n (%)	–	–	–	81 (86.2)	77 (81.9)	158 (84.0)
No, n (%)	–	–	–	13 (13.8)	17 (18.1)	30 (16.0)
Histology, n (%)						
Adenocarcinoma NOS	24 (8.1)	35 (11.7)	59 (9.9)	12 (10.9)	12 (10.7)	24 (10.8)
Clear cell	19 (6.4)	20 (6.7)	39 (6.5)	0 (0)	0 (0)	0 (0)
Dedifferentiated or undifferentiated	7 (2.3)	6 (2.0)	13 (2.2)	4 (3.6)	4 (3.6)	8 (3.6)
Endometrioid						
Grade 1	56 (18.8)	46 (15.4)	102 (17.1)	20 (18.2)	34 (30.4)	54 (24.3)
Grade 2	52 (17.4)	59 (19.7)	111 (18.6)	52 (47.3)	43 (38.4)	95 (42.8)
Grade 3	53 (17.8)	45 (15.1)	98 (16.4)	15 (13.6)	16 (14.3)	31 (14.0)
Mixed epithelial	6 (2.0)	10 (3.3)	16 (2.7)	3 (2.7)	2 (1.8)	5 (2.3)
Serous	81 (27.2)	77 (25.8)	158 (26.5)	4 (3.6)	1 (0.9)	5 (2.3)
Missing	0 (0)	1 (0.3)	1 (0.2)	0 (0)	0 (0)	0 (0)
FIGO stage at initial diagnosis						
I	105 (35.2)	104 (34.8)	209 (35.0)	45 (40.9)	57 (50.9)	102 (45.9)
II	25 (8.4)	25 (8.4)	50 (8.4)	13 (11.8)	13 (11.6)	26 (11.7)
III	57 (19.1)	47 (15.7)	104 (17.4)	18 (16.4)	11 (9.8)	29 (13.1)
IVA	11 (3.7)	5 (1.7)	16 (2.7)	2 (1.8)	4 (3.6)	6 (2.7)
IVB	99 (33.2)	118 (39.5)	217 (36.3)	32 (29.1)	27 (24.1)	59 (26.6)
Missing	1 (0.3)	0	1 (0.2)	0	0	0
Previous therapy, n (%)						
Chemotherapy	75 (25.2)	77 (25.8)	152 (25.5)	4 (3.6)	8 (7.1)	12 (5.4)
Radiotherapy	120 (40.3)	126 (42.1)	246 (41.2)	42 (38.2)	54 (48.2)	96 (43.2)
Surgery	269 (90.3)	253 (84.6)	522 (87.4)	97 (88.2)	104 (92.9)	201 (90.5)

Data cutoff date: August 18, 2023.

CPS, combined positive score; dMMR, mismatch repair-deficient; ECOG PS, Eastern Cooperative Oncology Group performance status; FIGO, International Federation of Gynecology and Obstetrics; ITT, intention to treat; NOS, not otherwise specified; PD-L1, programmed cell death ligand 1; pMMR mismatch repair-proficient.

[a]CPS was defined as the number of PD-L1–positive cells (tumor cells, lymphocytes, and macrophages) divided by the total number of tumor cells, multiplied by 100.

https://doi.org/10.1038/s41591-025-03566-1

Extended Data Table 2 | Investigator-Assessed Best Overall Response per RECIST v1.1 at Ad Hoc Analysis (ITT Population With Measurable Disease at Baseline)

	pMMR Population		dMMR Population	
	Pembrolizumab + Chemotherapy (n = 224)	**Placebo + Chemotherapy (n = 239)**	**Pembrolizumab + Chemotherapy (n = 95)**	**Placebo + Chemotherapy (n = 95)**
ORR, n (%) [95% CI]	162 (72.3) [66.0–78.1]	141 (59.0) [52.5–65.3]	78 (82.1) [72.9–89.2]	68 (71.6) [61.4–80.4]
Best overall response, n (%)				
CR	32 (14.3)	20 (8.4)	30 (31.6)	13 (13.7)
PR	130 (58.0)	121 (50.6)	48 (50.5)	55 (57.9)
SD	26 (11.6)	55 (23.0)	7 (7.4)	16 (16.8)
PD	16 (7.1)	19 (7.9)	4 (4.2)	3 (3.2)
Not evaluable[a]	2 (0.9)	3 (1.3)	0	0
No assessment[b]	18 (8.0)	21 (8.8)	6 (6.3)	8 (8.4)
TTR, median (range), mo	2.3 (1.9–19.6)	2.3 (1.0–7.1)	2.3 (1.6–11.6)	2.3 (2.0–14.5)
DOR, median (range), mo	8.1 (0.0+ to 40.9+)	6.4 (0.0+ to 28.3+)	NR (0.0+ to 41.8+)	4.8 (0.0+ to 42.2+)

Data cutoff date: August 18, 2023.
"+" indicates no PD by the time of last disease assessment.
CR, complete response; dMMR, mismatch repair-deficient; DOR, duration of response; ITT, intention to treat; NR, not reached; ORR, objective response rate; PD, progressive disease; pMMR mismatch repair-proficient; PR, partial response; RECIST, Response Evaluation Criteria in Solid Tumors; SD, stable disease; TTR, time to response.
[a]Postbaseline assessment available but not evaluable.
[b]No postbaseline assessment available for response evaluation.

https://doi.org/10.1038/s41591-025-03566-1

Extended Data Table 3 | Investigator-Assessed Best Overall Response Per RECIST v1.1 at Interim Analysis (ITT Population With Measurable Disease at Baseline)

	pMMR Population		dMMR Population	
	Pembrolizumab + Chemotherapy (n = 220)	**Placebo + Chemotherapy (n = 235)**	**Pembrolizumab + Chemotherapy (n = 95)**	**Placebo + Chemotherapy (n = 95)**
ORR, n (%) [95% CI]	135 (61.4) [54.6–67.8]	121 (51.5) [44.9–58.0]	74 (77.9) [68.2–85.8]	66 (69.5) [59.2–78.5]
Best overall response, n (%)				
CR	24 (10.9)	16 (6.8)	27 (28.4)	11 (11.6)
PR	111 (50.5)	105 (44.7)	47 (49.5)	55 (57.9)
SD	29 (13.2)	52 (22.1)	10 (10.5)	17 (17.9)
PD	12 (5.5)	19 (8.1)	5 (5.3)	3 (3.2)
Not evaluable[a]	2 (0.9)	2 (0.9)	0	1 (1.1)
No assessment[b]	42 (19.1)	41 (17.4)	6 (6.3)	8 (8.4)
TTR, median (range), mo	2.3 (1.9–19.6)	2.3 (1.0–7.1)	2.3 (1.3–11.5)	2.2 (2.0–6.2)
DOR, median (range), mo	7.1 (0.0+ to 32.8+)	6.4 (0.0+ to 20.1+)	NR (0.0+ to 33.0+)	4.4 (0.0+ to 32.8+)

Data cutoff dates: December 6, 2022 (pMMR); December 16, 2022 (dMMR).
"+" indicates no PD by the time of last disease assessment.
CR, complete response; dMMR, mismatch repair-deficient; DOR, duration of response; ITT, intention to treat; NR, not reached; ORR, objective response rate; PD, progressive disease; pMMR mismatch repair-proficient; PR, partial response; RECIST, Response Evaluation Criteria in Solid Tumors; SD, stable disease; TTR, time to response.
[a]Postbaseline assessment available but not evaluable.
[b]No postbaseline assessment available for response evaluation.

nature portfolio

Corresponding author(s): Ramez N. Eskander

Last updated by author(s): Jan 17, 2025

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	N/A
Data analysis	N/A

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

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Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	The manuscript reports on patients with endometrial cancer.
Reporting on race, ethnicity, or other socially relevant groupings	Race and ethnicity data are reported in Table 1 and Extended Data Table 1. To control for confounding variables, patients were randomized to treatment. The patients, investigators and the sponsor personnel or delegate(s) who were involved in the treatment administration or clinical evaluation of the patients were unaware of the group assignments. The chemotherapy agents were open-label.
Population characteristics	This information is provided in Table 1 and Extended Data Table 1.
Recruitment	Patient enrollment for this 2-step (registration and randomization) trial was facilitated using the Oncology Patient Enrollment Network (OPEN). All site staff used OPEN to enroll patients to this study. It is integrated with the CTSU Enterprise System for regulatory and roster data and, upon enrollment, initializes the patient in the Rave database. Prior to accessing OPEN, site staff verified the following:All eligibility criteria have been met within the protocol stated timeframes; All patients have signed an appropriate consent form and HIPAA authorization form (if applicable).
Ethics oversight	The study was conducted in accordance with the International Council for Harmonisation Good Clinical Practice guidelines and ethical principles that have their origin in the Declaration of Helsinki. The protocol was approved by an ethics body at each study site (see Data Supplement for a list of study sites). Patients provided written informed consent. Patients were not compensated for their participation in the study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We determined that the enrollment of 810 patients — 220 in the dMMR cohort and 590 in the pMMR cohort — would provide the trial with 85% power to detect a relative reduction of 40% in the risk of progression or death as compared with chemotherapy alone in the dMMR cohort after the occurrence of 168 primary-outcome events and 90% power to detect a relative reduction of 30% in the pMMR cohort after the occurrence of 394 events.
Data exclusions	Patients with indeterminate central MMR status were excluded as per protocol. Nine patients in the pMMR population were randomized after the December 6, 2022 DCO and therefore were not included in both the current and previous PFS and OS analyses using the IA DCOs, but are included in the ad hoc analyses. This information is detailed in the Statistical Analyses section of the manuscript
Replication	N/A
Randomization	Patients were randomized 1:1 to treatment. Randomization was stratified within the following groups: Mismatch repair deficient (dMMR) (yes/no); Performance Status (0 or 1 vs 2); Prior chemotherapy (yes/no)
Blinding	This is a double-blind trial; therefore, the patients, the investigators and the sponsor personnel or delegate(s) involved in the treatment administration or clinical evaluation of the patients were unaware of the group assignments. The chemotherapy agents were open-label. The sponsor, investigator and patient did not know whether the treatment administered contained pembrolizumab or saline placebo. Each study site had an unblinded pharmacist that obtained each patient's study identification number, and drug assignment and then prepare the solutions for infusion. The unblinded pharmacist provided the investigative staff with the ready-to-use blinded pembrolizumab/saline infusion solutions, packaged identically in order to maintain the blinding, for administration at scheduled infusion visits.

Behavioural & social sciences study design

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Study description	*Briefly describe the study type including whether data are quantitative, qualitative, or mixed-methods (e.g. qualitative cross-sectional, quantitative experimental, mixed-methods case study).*
Research sample	*State the research sample (e.g. Harvard university undergraduates, villagers in rural India) and provide relevant demographic information (e.g. age, sex) and indicate whether the sample is representative. Provide a rationale for the study sample chosen. For studies involving existing datasets, please describe the dataset and source.*
Sampling strategy	*Describe the sampling procedure (e.g. random, snowball, stratified, convenience). Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient. For qualitative data, please indicate whether data saturation was considered, and what criteria were used to decide that no further sampling was needed.*
Data collection	*Provide details about the data collection procedure, including the instruments or devices used to record the data (e.g. pen and paper, computer, eye tracker, video or audio equipment) whether anyone was present besides the participant(s) and the researcher, and whether the researcher was blind to experimental condition and/or the study hypothesis during data collection.*
Timing	*Indicate the start and stop dates of data collection. If there is a gap between collection periods, state the dates for each sample cohort.*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, provide the exact number of exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Non-participation	*State how many participants dropped out/declined participation and the reason(s) given OR provide response rate OR state that no participants dropped out/declined participation.*
Randomization	*If participants were not allocated into experimental groups, state so OR describe how participants were allocated to groups, and if allocation was not random, describe how covariates were controlled.*

Ecological, evolutionary & environmental sciences study design

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Study description	*Briefly describe the study. For quantitative data include treatment factors and interactions, design structure (e.g. factorial, nested, hierarchical), nature and number of experimental units and replicates.*
Research sample	*Describe the research sample (e.g. a group of tagged Passer domesticus, all Stenocereus thurberi within Organ Pipe Cactus National Monument), and provide a rationale for the sample choice. When relevant, describe the organism taxa, source, sex, age range and any manipulations. State what population the sample is meant to represent when applicable. For studies involving existing datasets, describe the data and its source.*
Sampling strategy	*Note the sampling procedure. Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.*
Data collection	*Describe the data collection procedure, including who recorded the data and how.*
Timing and spatial scale	*Indicate the start and stop dates of data collection, noting the frequency and periodicity of sampling and providing a rationale for these choices. If there is a gap between collection periods, state the dates for each sample cohort. Specify the spatial scale from which the data are taken*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
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Did the study involve field work? ☐ Yes ☐ No

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Field conditions	*Describe the study conditions for field work, providing relevant parameters (e.g. temperature, rainfall).*
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Materials & experimental systems

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☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	*Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.*
Validation	*Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.*

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)	*State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.*
Authentication	*Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.*
Mycoplasma contamination	*Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.*
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If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.

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Ethics oversight

Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

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Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals

For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.

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Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.

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Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.

Field-collected samples

For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.

Ethics oversight

Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration

ClinicalTrials.gov, NCT03914612

Study protocol

NRG GY018

Data collection

Data collection for this study exclusively through Medidata Rave

Outcomes

Results for the primary endpoint, PFS per RECIST v1.1 as assessed by the investigator in the pMMR and dMMR populations, were previously reported (Eskander N Engl J Med 2023). Secondary endpoints reported include concordance between local and central MMR IHC testing; PFS per RECIST v1.1 as assessed by BICR; OS; ORR, best overall response (BOR), and DOR per RECIST v1.1 as assessed by the investigator; and safety. Exploratory endpoints presented include the association between PD-L1 CPS with MMR status and subgroup analyses of investigator-assessed PFS per RECIST v1.1. Secondary and exploratory endpoints were assessed separately in the pMMR and dMMR populations except for safety (combined population) and subgroup analyses of investigator-assessed PFS based on methylation status (dMMR population only).
Efficacy analyses at IA (PFS per investigator assessment including in subgroups, PFS by BICR, and OS) were conducted in the intention-to-treat population. Because some patients had no postbaseline assessment available for response evaluation at the December 2022 DCOs, an ad hoc analysis of ORR, BOR, and DOR was performed using a later DCO of August 18, 2023. Patients had to have measurable disease at baseline to be included in the analyses of objective responses. OS in the intention-to-treat population, along with sensitivity analysis for poststudy subsequent anticancer therapy, and safety in all randomized and treated patients were also assessed as part of the ad hoc analysis. Statistical methods for the confounding effect of subsequent anticancer therapy on OS are described in the Data Supplement.

Dual use research of concern

Policy information about dual use research of concern

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes	
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☒	☐	Crops and/or livestock
☒	☐	Ecosystems
☒	☐	Any other significant area

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Does the work involve any of these experiments of concern:

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☒	☐	Increase transmissibility of a pathogen
☒	☐	Alter the host range of a pathogen
☒	☐	Enable evasion of diagnostic/detection modalities
☒	☐	Enable the weaponization of a biological agent or toxin
☒	☐	Any other potentially harmful combination of experiments and agents

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

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Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.

ChIP-seq

Data deposition

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☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links
May remain private before publication.

For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.

Files in database submission

Provide a list of all files available in the database submission.

Genome browser session
(e.g. UCSC)

Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.

Methodology

Replicates

Describe the experimental replicates, specifying number, type and replicate agreement.

Sequencing depth

Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.

Antibodies

Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.

Peak calling parameters

Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.

Data quality	*Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.*
Software	*Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.*

Flow Cytometry

Plots

Confirm that:

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☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☐ All plots are contour plots with outliers or pseudocolor plots.

☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	*Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.*
Instrument	*Identify the instrument used for data collection, specifying make and model number.*
Software	*Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.*
Cell population abundance	*Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.*
Gating strategy	*Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.*

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type	*Indicate task or resting state; event-related or block design.*
Design specifications	*Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.*
Behavioral performance measures	*State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).*

Acquisition

Imaging type(s)	*Specify: functional, structural, diffusion, perfusion.*
Field strength	*Specify in Tesla*
Sequence & imaging parameters	*Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.*
Area of acquisition	*State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.*

Diffusion MRI ☐ Used ☐ Not used

Preprocessing

Preprocessing software	*Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).*
Normalization	*If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.*

Normalization template

Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.

Noise and artifact removal

Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).

Volume censoring

Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.

Statistical modeling & inference

Model type and settings

Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).

Effect(s) tested

Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.

(See Eklund et al. 2016)

Correction

Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).

Models & analysis

n/a	Involved in the study
☐	☐ Functional and/or effective connectivity
☐	☐ Graph analysis
☐	☐ Multivariate modeling or predictive analysis

Functional and/or effective connectivity

Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).

Graph analysis

Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).

Multivariate modeling and predictive analysis

Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.